The Mechanism of Thermalization in the Bost–Connes System

Shao-Wu Hu

Shanghai, China

forest@xindong.com

ORCID: 0009-0009-8300-4691

Abstract

Thermalization brings matter toward equilibrium through sustained exchange, while information about the exchange processes remains in the full joint state. Starting from the dynamical cycle established on a finite reduction of the Bost–Connes system, we study the relation between thermal equilibrium, relaxation rates, and local irreversibility. At a fixed finite temperature, with independently refreshed environments, positive exchange shares, and same-temperature restoration as specified here, the full Gibbs state is the unique attracting state of the cycle. The occupation spectrum and coherence factors determine the relaxation rates of different observables and yield a controlled weak-exchange continuous limit on the restored family of states. Energy changes include both thermal exchange and background restoration. This gives conditions for tracking thermal states as the temperature changes, together with the late-time cooling law and its asymptotic corrections for a specified zero-temperature exit process.

The same joint realization determines the environment outputs and entropy balance. The decrease in system relative entropy characterizes the approach to equilibrium, while changes in auxiliary preparations enter the complete correlation and energy balance. The auxiliaries can continue to change even when the system is at equilibrium. We also obtain a return process in which the same interaction acts again on previously used auxiliaries. The endpoint, rate, and time direction of thermalization are thus determined by the equilibrium conditions, actual decay modes, and environmental boundary conditions of one process. They can be tested jointly through occupation, coherence, and energy-flow readings constrained by common parameters.

Keywords: Bost–Connes system; thermalization; quantum dynamics; relaxation; entropy production; environment records

Contents

1 Physical Picture of Thermalization **3**
- 1.1 Energy Exchange . . . 3
- 1.2 Evolution of the Distribution . . . 5
- 1.3 Thermal Equilibrium . . . 6

2 Energy Exchange and Thermal Equilibrium **8**
- 2.1 Exchange between Matter and the Environment . . . 8
- 2.2 Occupation Balance . . . 9
- 2.3 The Thermal Equilibrium Distribution . . . 11
- 2.4 Establishment of Equilibrium . . . 13

3 Relaxation Time Scales **14**
- 3.1 Decay of Occupation Deviations . . . 14
- 3.2 Energy and Mode Composition . . . 16
- 3.3 Coherence Decay . . . 17
- 3.4 Continuous Relaxation under Weak Exchange . . . 18
- 3.5 Slow Modes and Thermalization Time . . . 20

4 Cooling and Temperature Evolution **22**
- 4.1 Temperature and Energy Distribution . . . 22
- 4.2 Thermalization at Changing Temperature . . . 23
- 4.3 Energy Flow and Cooling Rate . . . 26
- 4.4 Low-Temperature Cooling . . . 27
- 4.5 Corrections to the Cooling Curve . . . 28

5 Irreversibility of Thermalization **31**
- 5.1 Exchange Records . . . 31
- 5.2 Entropy and Thermal Equilibrium . . . 32
- 5.3 Entropy Balance of the Environment . . . 33
- 5.4 Forward and Reverse Processes and the Direction of Time . . . 36

6 Tests of the Thermalization Mechanism **39**
- 6.1 Exchange Statistics . . . 39
- 6.2 Joint Tests of Occupations and Coherence . . . 40
- 6.3 Cooling and Entropy Balance . . . 42
- 6.4 Environment Refreshing and Return . . . 43

7 Conclusion **46**

References **50**

§1 Physical Picture of Thermalization

Sustained exchange between matter and its environment changes energy occupations and carries distinctions between processes into the environment outputs. In *Dynamics of the Bost–Connes State Space* [1], we established a complete cycle comprising free evolution, thermal exchange, coherent retrieval, and background restoration. These processes occur continuously in the joint space of the system and auxiliary units. Taking the partial trace over the auxiliary outputs of the cycle gives the complete system evolution

$$\boxed{\mathcal{F}(\rho) = \mathrm{Tr}_{\mathcal{E}}\left[\mathbb{U}_{\mathrm{tot}}(\rho \otimes \eta_{\mathrm{full}})\mathbb{U}_{\mathrm{tot}}^{\dagger}\right].} \tag{1.1}$$

Here, $\mathbb{U}_{\mathrm{tot}}$ is the complete joint action, η_{full} is the auxiliary preparation for the cycle, and $\mathcal{E}$ includes the thermal environment and the auxiliaries responsible for return and restoration. The final system state retains all contributions from matter return, exit, and renewed formation; the auxiliary outputs carry the corresponding energy changes and correlations.

Reference [1] gives conditions under which this cycle preserves the full Gibbs state and brings different initial states toward the same thermal distribution.

§1.1 Energy Exchange

Energy exchange in this cycle acts jointly on matter and the environment. Suppose that the system moves from a state of energy E_a to one of energy E_b, while the corresponding environment energy changes from $\varepsilon_{\mathrm{in}}$ to $\varepsilon_{\mathrm{out}}$. An energy-conserving channel satisfies

$$E_b - E_a = \varepsilon_{\mathrm{in}} - \varepsilon_{\mathrm{out}}. \tag{1.2}$$

The energy gained by the system comes from the environment, and the reverse transfer releases energy to the environment. Allowed exchanges must also satisfy the corresponding charge conservation laws. Their actual amplitudes are determined by the joint matrix elements and interaction durations [1,2]. The same joint action includes both directions. The incoming environment preparation and the channel action together determine their transfer shares in a cycle.

The final joint state generally contains several exchange branches. No exchange, absorption, and emission can correspond to different environment outputs. When the full outputs distinguish these branches, tracing out the environment changes the corresponding system interference. Different energy levels can also produce the same outgoing reading, so whether an exchange distinguishes a particular coherence component must be determined from the actual joint output. Both energy transfer and coherence change follow from this output.

The share transferred from matter to the background also participates in subsequent restoration and renewed formation. The background here denotes the shell-average component inside the system; it is accounted for separately from the incoming external environment in the joint state. The thermal environment compensates energy and charge changes during exchange, and its outgoing state carries the corresponding exchange information. The restoration auxiliary rearranges the background distribution and preserves distinctions between its input and output

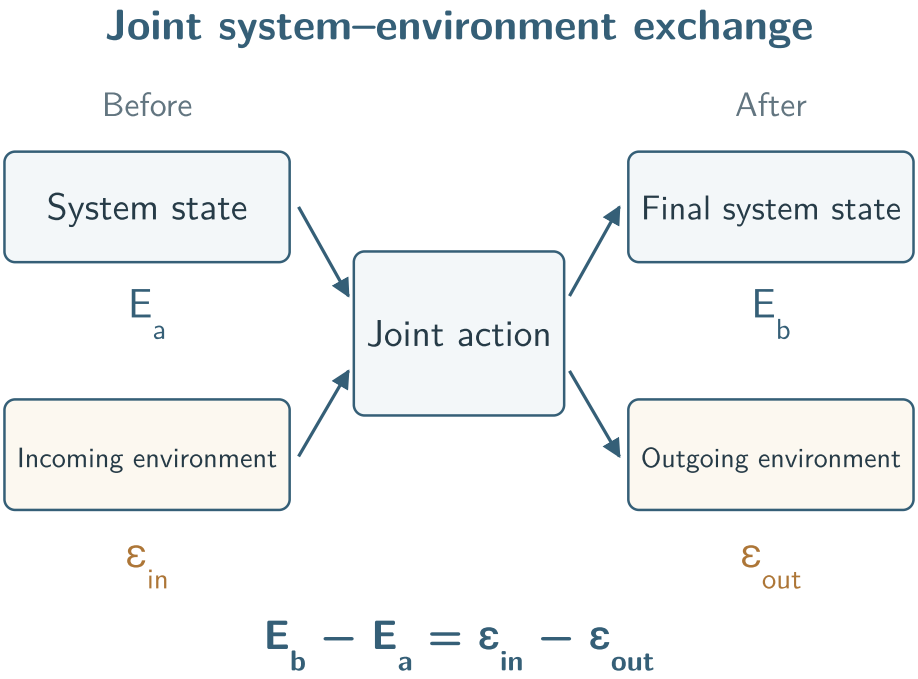


Figure 1-1. **A local exchange.** The system and incoming environment participate jointly in the interaction. The final state includes both the system change and the environment output. Forward and reverse exchanges balance energy term by term, while the allowed branches are still described by joint amplitudes. Arrows indicate relations between the states before and after the interaction. Whether particular branches form distinguishable records is determined by the full environment output.

in its own output. A physical environment carrier must realize these joint matrix elements and conservation conditions; the forward-to-reverse exchange ratio alone does not determine its particle species. An auxiliary involved in coherent retrieval must be retained until return is complete. Outputs that have departed are treated according to the actual boundary conditions. One cycle includes these actions and their energy balance [1], and its final state sets the starting point for the next cycle.

§1.2 Evolution of the Distribution

We use the finite reduction $R_K = \mathbb{Z}/p^K\mathbb{Z}$ and state space $\mathcal{H}_K = \ell^2(R_K)$, where p is a fixed odd prime and $K \geq 1$ is the truncation depth. Shell j consists of elements of valuation j, and the terminal shell contains only the zero element. The shell dimensions $d_j = |S_j|$ are

$$d_j = \begin{cases} (p-1)p^{K-j-1}, & 0 \leq j < K, \\ 1, & j = K, \end{cases} \qquad \sum_{j=0}^{K} d_j = p^K. \tag{1.3}$$

Let ρ_n be the system state after the nth complete cycle. We use the established thermal-exchange energy $H_K = \sum_j E_j P_j$, where P_j projects onto shell j and $E_j = j \log p$. Throughout, $\hbar = k_{\mathrm{B}} = 1$, and log denotes the natural logarithm. Shell occupations and mean energy are

$$q_j(n) = \mathrm{Tr}(P_j \rho_n), \qquad \sum_{j=0}^{K} q_j(n) = 1,$$
$$E_S(n) = \sum_{j=0}^{K} E_j q_j(n). \tag{1.4}$$

The quantity $q_j(n)$ is the probability of measuring that energy shell after n cycles under the same preparation. A shell can still contain different matter modes and their coherences, so shell occupations are only one class of readings of the full state. Shell labels here describe valuation and energy levels; spatial displacement of matter is described by the corresponding position degrees of freedom [2].

With the joint action and environment preparation fixed for all cycles, the complete map $\mathcal{F}$ introduced above is applied repeatedly. Each fresh environment and restoration auxiliary is independent of the joint state of the system and all previous environments, and old outputs no longer participate. This description through successive auxiliary units originates in stepwise relaxation models [3] and has been developed in finite-dimensional thermalization models [4] and repeated-interaction theory [5]. Under these conditions,

$$\boxed{\rho_{n+1} = \mathcal{F}(\rho_n), \qquad \rho_n = \mathcal{F}^n(\rho_0).} \tag{1.5}$$

The same map $\mathcal{F}$ acts repeatedly on a changing state. The actual weight of an exchange branch depends on both the channel and the current occupation. A larger occupation of a mode in one cycle changes the total share of exchanges originating

from that mode in the next cycle. The combined channel balances change $q_j(n)$ and hence the distribution on which subsequent exchanges act.

This iteration retains all branches of return, exit, and renewed formation. Keeping only histories that have never exited would require renormalizing the conditional sample, whose distribution generally differs from that of the full ensemble considered here [1].

§1.3 Thermal Equilibrium

A stationary distribution is unchanged by one cycle. To represent thermal equilibrium at inverse temperature β, it must also be compatible with the actual exchange energy. For H_K, the Gibbs state on the full space is

$$\rho_\beta = \frac{e^{-\beta H_K}}{Z_K}, \qquad Z_K = \sum_{j=0}^{K} |S_j| e^{-\beta E_j}. \tag{1.6}$$

Every independent state at the same energy has the same thermal weight, so the total occupation of that shell is

$$q_j^{\mathrm{eq}} = \frac{|S_j| e^{-\beta E_j}}{Z_K}. \tag{1.7}$$

Energy and degeneracy jointly determine the full thermal distribution. A conditional state restricted to the subspace containing one average direction per shell has a partition function computed with the dimensions of those directions. It and the full shell occupations refer to distinct statistical objects.

Forward and reverse exchanges at the same temperature preserve Gibbs weights, and background restoration must be compatible with that temperature as well. If the complete cycle preserves ρ_β, and positive exchange and restoration suffice to remove deviations between initial states, then

$$\mathcal{F}(\rho_\beta) = \rho_\beta, \qquad \mathcal{F}^n(\rho_0) \longrightarrow \rho_\beta. \tag{1.8}$$

The first relation expresses preservation of the thermal state; the second describes convergence toward it. Together they constitute thermalization [1]. At equilibrium, forward and reverse probability flows are equal in channels satisfying detailed balance, while microscopic exchange can continue. Stationarity of the system distribution does not require every environment auxiliary to remain unchanged.

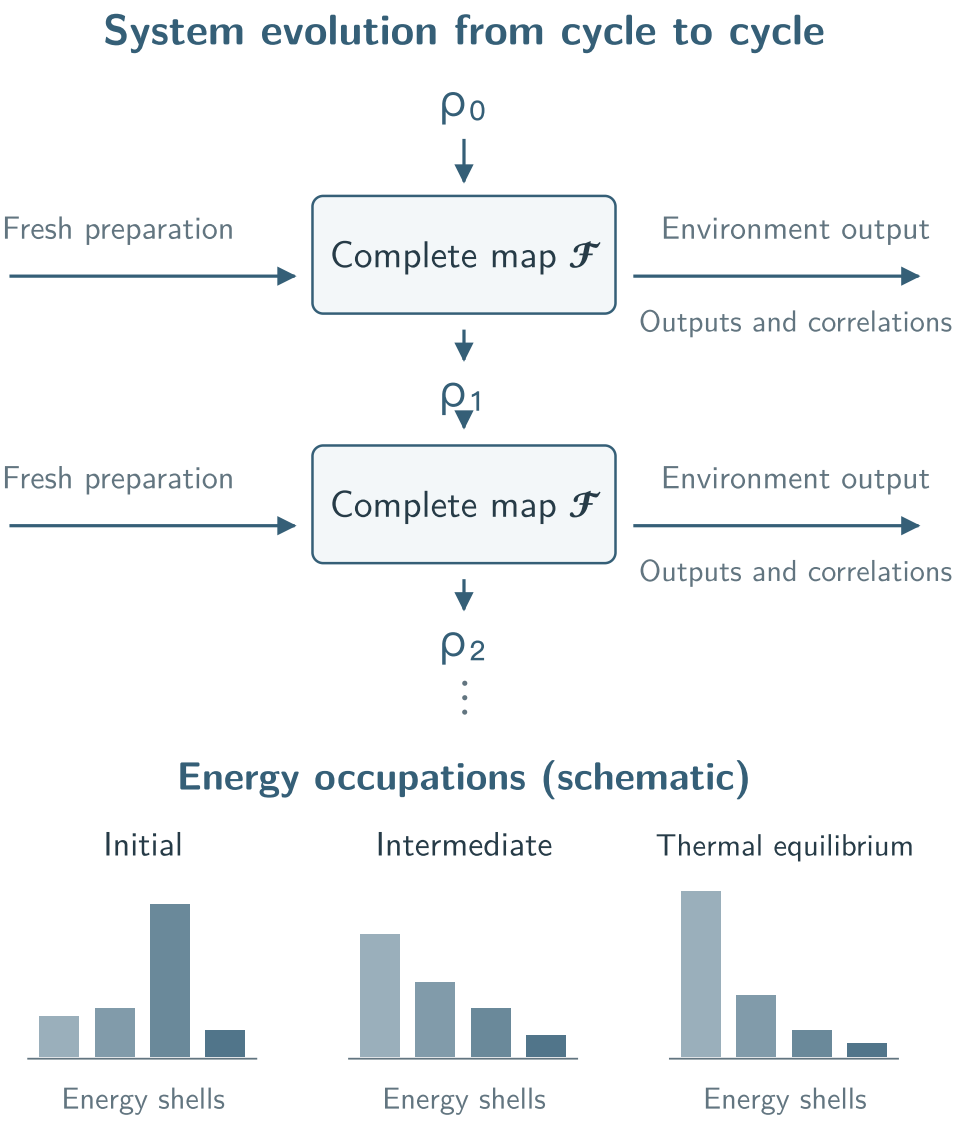


Figure 1-2. **From local exchange to repeated evolution.** The system state continues from one cycle to the next along the central line. Each cycle receives a prescribed fresh environment and retains its outputs. A complete cycle includes exchange, return, and restoration. The bar charts schematically show changes in energy occupations from an initial deviation to thermal equilibrium. They are not calculations for specific parameters and do not require every shell occupation to vary monotonically.

§2 Energy Exchange and Thermal Equilibrium

Energy exchange first changes the states participating in a channel and then, through repeated action, changes the overall occupations. The relative shares of forward and reverse processes are constrained by the environment preparation, while the net flow depends on these shares and the current occupations. For the complete cycle introduced above, choosing propagation stages that do not additionally mix matter modes allows the exchange and occupation balances to be calculated term by term, revealing how thermal equilibrium is established. The energy used throughout is H_K; return phases and background restoration belong to the same cycle.

§2.1 Exchange between Matter and the Environment

We represent the selected thermal cycle in terms of directional excitations and the shell-average background, decomposing the finite state space orthogonally into the directional excitation space $\mathcal{M}$ and the shell-average subspace $\mathcal{C}$. Hereafter, matter modes are the directional excitations tracked by this cycle; collective matter patterns can also have other representations [2]. Denote these matter modes by $|\nu\rangle$ and the normalized average mode of shell a by $|e_a\rangle$. Together they form an orthonormal basis, with projections P_ν and P_{e_a} and energies E_ν and E_a, respectively. Allowed exchanges between these modes are supplied by the established joint construction [1].

For a selected channel (a, ν), the system-side matrix unit is

$$F_{a\nu} = |e_a\rangle\langle\nu|, \qquad [H_K, F_{a\nu}] = (E_a - E_\nu)F_{a\nu}. \tag{2.1}$$

Write $\Delta_{a\nu} = E_a - E_\nu$. Assign orthogonal environment states $|u\rangle, |v\rangle$ to this channel, with energies and charges that balance at the two joint endpoints. In particular,

$$E_\nu + \varepsilon_u = E_a + \varepsilon_v, \qquad \varepsilon_u - \varepsilon_v = \Delta_{a\nu}. \tag{2.2}$$

Let $L_{a\nu} = F_{a\nu} \otimes |v\rangle\langle u|$. The joint exchange is $W_{a\nu} = \exp[\alpha_{a\nu}(L_{a\nu} - L_{a\nu}^\dagger)]$, where $\alpha_{a\nu}$ is the actual exchange angle. The generator is anti-Hermitian on the span of the two joint endpoints and vanishes on its orthogonal complement. Its exponential gives

$$\begin{aligned} W_{a\nu}|\nu, u\rangle &= \cos\alpha_{a\nu}|\nu, u\rangle + \sin\alpha_{a\nu}|e_a, v\rangle, \\ W_{a\nu}|e_a, v\rangle &= \cos\alpha_{a\nu}|e_a, v\rangle - \sin\alpha_{a\nu}|\nu, u\rangle. \end{aligned} \tag{2.3}$$

The first relation gives the branches of matter return and transfer into the background; the second gives background continuation and matter formation. Both transfers have the same exchange angle, and the environment carries the opposite energy and charge changes.

The thermal environment is prepared in $t_u|u\rangle\langle u| + t_v|v\rangle\langle v|$ within this block. At a common inverse temperature β,

$$\begin{gathered} t_u = \frac{1}{1 + e^{\beta\Delta_{a\nu}}}, \qquad t_v = \frac{e^{\beta\Delta_{a\nu}}}{1 + e^{\beta\Delta_{a\nu}}}, \\ \frac{t_u}{t_v} = e^{-\beta\Delta_{a\nu}}. \end{gathered} \tag{2.4}$$

Let $a_{a\nu} > 0$ be the preparation weight of the selected channel, and retain a weight $a_0 = 1 - \sum_{a,\nu} a_{a\nu} > 0$ for no thermal exchange. Different channels occupy orthogonal environment blocks, whose weights belong to the actual preparation. Squared norms of the exchange branches give the exit and reverse-formation shares in one cycle:

$$\boxed{\begin{aligned} \epsilon_{a\nu} &= a_{a\nu} t_u \sin^2 \alpha_{a\nu}, \\ f_{\nu a} &= a_{a\nu} t_v \sin^2 \alpha_{a\nu}. \end{aligned}} \tag{2.5}$$

Every positive exchange channel therefore satisfies

$$\frac{\epsilon_{a\nu}}{f_{\nu a}} = e^{-\beta(E_a - E_\nu)}. \tag{2.6}$$

Temperature specifies the ratio of the forward and reverse shares; channel weights and exchange angles set their absolute values. All shares below are counted per cycle. The corresponding mean counting rates require division by the total elapsed duration, including exchange, return, restoration, and the necessary preparation operations.

§2.2 Occupation Balance

After exchange, the returning component remains in the matter-mode space, while the background component is restored with fixed shares g_a. Write

$$\sigma_g^{\mathrm{sh}} = \sum_a g_a P_{e_a}, \qquad g_a > 0, \quad \sum_a g_a = 1. \tag{2.7}$$

Coherent retrieval first returns the auxiliary to its initial preparation. The same auxiliary then participates in background restoration, preserving distinctions in the incoming background through orthogonal output labels. After tracing over the full outputs of the thermal environment and this auxiliary, the first-cycle final state of any initial state has the form [1]

$$\begin{aligned} \rho_n &= X_n + y_n \sigma_g^{\mathrm{sh}}, \qquad n \geq 1, \\ X_n &\geq 0, \quad \operatorname{Tr} X_n + y_n = 1. \end{aligned} \tag{2.8}$$

Here X_n is supported on $\mathcal{M}$ and contains both mode occupations and intermode coherences; y_n is the total background share. Every subsequent cycle preserves this form. If the initial state already has this form, the recursion starts at $n = 0$.

Let $x_\nu(n) = \langle \nu | X_n | \nu \rangle$. The background endpoint occupation before exchange is $y_n g_a$, so the net exit share of a channel is

$$\boxed{J_{a\nu}(n) = \epsilon_{a\nu} x_\nu(n) - f_{\nu a} y_n g_a.} \tag{2.9}$$

A positive value means net transfer from matter to the background, and a negative value means that reverse formation dominates. The occupations after thermal exchange are

$$\widetilde{x}_\nu = x_\nu - \sum_a J_{a\nu}, \qquad \widetilde{z}_a = y_n g_a + \sum_\nu J_{a\nu}. \tag{2.10}$$

Background restoration preserves the total share $\widetilde{y} = \sum_a \widetilde{z}_a$ and redistributes it as $\widetilde{y} g_a$. Return phases do not change mode occupations, so a complete cycle obeys

$$\boxed{\begin{aligned} x_\nu(n+1) - x_\nu(n) &= -\sum_a J_{a\nu}(n), \\ y_{n+1} - y_n &= \sum_{a,\nu} J_{a\nu}(n). \end{aligned}} \tag{2.11}$$

Summing over all channels cancels inflows against outflows and preserves total probability. The updated occupations determine the net flows in the next cycle.

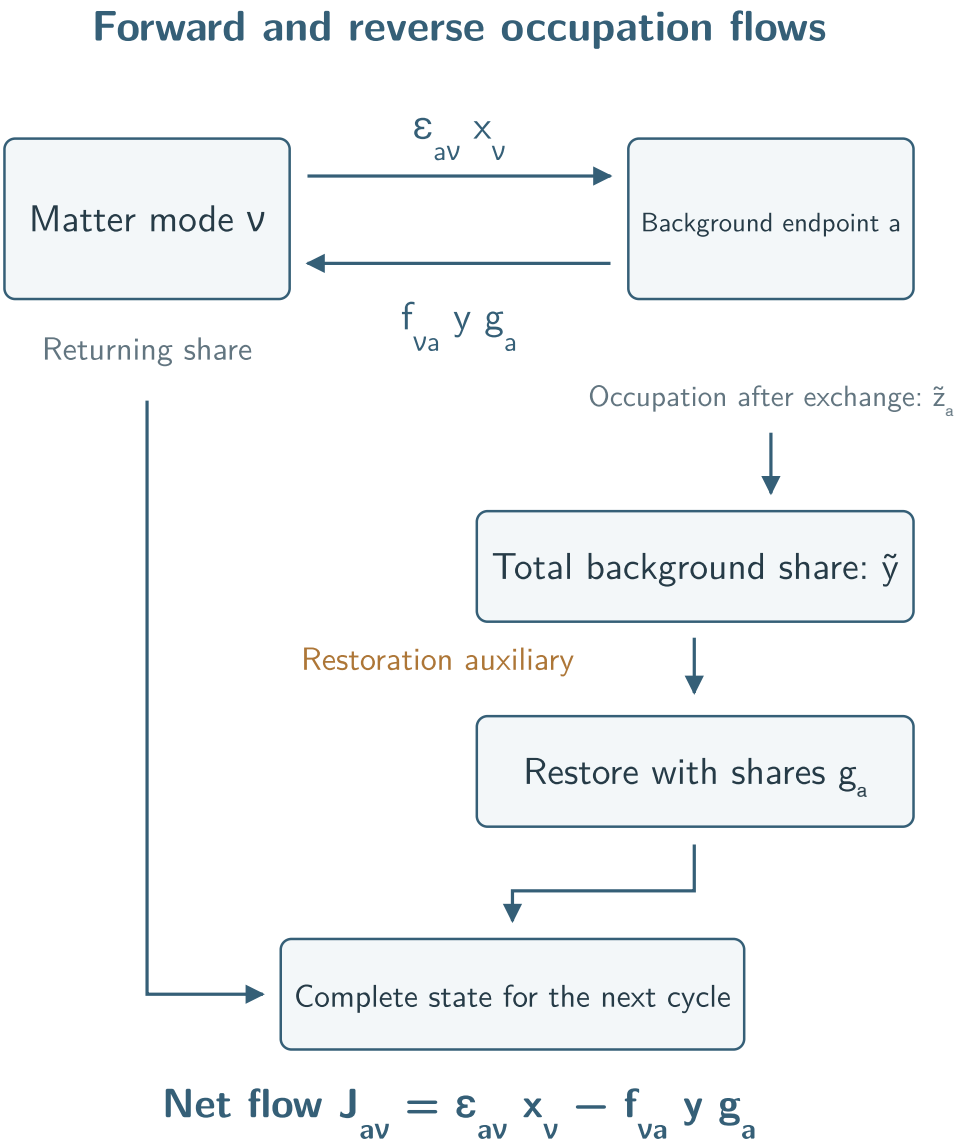


Figure 2-1. **Exchange flows and background restoration.** Forward arrows represent matter transfer to the background, and reverse arrows represent matter formation from the background. Their difference is the net flow. The background endpoint occupations after thermal exchange are summed and then restored according to the fixed preparation. Restoration has its own auxiliary output and energy balance; the returning matter component continues into the next cycle.

Summing the channels, define $\epsilon_\nu = \sum_a \epsilon_{a\nu}$, $b_\nu(g) = \sum_a g_a f_{\nu a}$, $B_g = \sum_\nu b_\nu(g) P_\nu$, and $f_\Sigma = \operatorname{Tr} B_g$. Denoting the matter-return map by $\mathcal{Q}$, the full state satisfies the

closed recursion

$$\begin{aligned} X_{n+1} &= \mathcal{Q}(X_n) + y_n B_g, \\ y_{n+1} &= \mathrm{Tr}[X_n - \mathcal{Q}(X_n)] + (1 - f_\Sigma) y_n. \end{aligned} \tag{2.12}$$

The first term retains returning matter already present, and the second adds matter formed during the current cycle. The occupation equations are the diagonal entries of this matrix recursion; intermode coherences continue to evolve under $\mathcal{Q}$. Expanding cycle by cycle also separates the original component from those formed in later cycles:

$$X_n = \mathcal{Q}^{n-1}(X_1) + \sum_{m=1}^{n-1} y_m \mathcal{Q}^{n-1-m}(B_g). \tag{2.13}$$

The overall occupation can therefore persist for long times even though each uninterrupted history has a probability of exit.

The same process gives the energy balance. Taking energy gained by the system as positive, thermal exchange and background restoration contribute, respectively,

$$\begin{aligned} Q_T &= \sum_{a,\nu} (E_a - E_\nu) J_{a\nu}, \\ Q_R &= \widetilde{y}\, \overline{E}_g - \sum_a E_a \widetilde{z}_a, \\ \overline{E}_g &= \sum_a g_a E_a. \end{aligned} \tag{2.14}$$

The first relation follows from the energy difference in each exchange channel, and the second from the background energy change during restoration. The corresponding auxiliaries carry the opposite changes. The auxiliary used for coherent retrieval returns to its preparation at the end of that stage, and the selected propagation also preserves H_K. Hence,

$$\boxed{E_S(n+1) - E_S(n) = Q_T(n) + Q_R(n).} \tag{2.15}$$

Energy changes over several cycles are obtained by summing these per-cycle balances. Both whether energy has balanced and whether individual modes are still being redistributed can be tested within the same full state.

§2.3 The Thermal Equilibrium Distribution

At stationarity, exit from each matter mode balances formation from the background. If every $\epsilon_\nu > 0$, the occupation equations give

$$x_\nu^* \epsilon_\nu = y_* b_\nu(g). \tag{2.16}$$

Using $\sum_\nu x_\nu^* + y_* = 1$ then yields

$$y_* = \left(1 + \sum_\nu \frac{b_\nu(g)}{\epsilon_\nu}\right)^{-1}, \qquad x_\nu^* = y_* \frac{b_\nu(g)}{\epsilon_\nu}. \tag{2.17}$$

These are the stationary occupations of the selected modes. Background restoration enters through $b_\nu(g)$. Specifying the forward-to-reverse thermal-exchange ratio alone does not yet specify the endpoint of the complete process.

Choose restoration shares at the same temperature as the thermal environment:

$$g_a = \pi_a = \frac{e^{-\beta E_a}}{Z_{\mathcal{C}}}, \qquad Z_{\mathcal{C}} = \sum_a e^{-\beta E_a}. \tag{2.18}$$

The exchange ratio gives, channel by channel,

$$\pi_a f_{\nu a} = \frac{e^{-\beta E_\nu}}{Z_{\mathcal{C}}}\epsilon_{a\nu}, \qquad b_\nu(\pi) = \frac{e^{-\beta E_\nu}}{Z_{\mathcal{C}}}\epsilon_\nu. \tag{2.19}$$

The matter and shell-average modes together form a complete orthonormal basis, so

$$Z_K = Z_{\mathcal{C}} + \sum_\nu e^{-\beta E_\nu} = \sum_j |S_j| e^{-\beta E_j}. \tag{2.20}$$

Substitution into the stationary solution gives

$$x^*_\nu = \frac{e^{-\beta E_\nu}}{Z_K}, \qquad y_* = \frac{Z_{\mathcal{C}}}{Z_K}. \tag{2.21}$$

These are precisely the diagonal occupations of the full Gibbs state. Since $y_*\pi_a = e^{-\beta E_a}/Z_K$, the channelwise relation also gives $J_{a\nu} = 0$ for every channel. Forward and reverse events still have nonzero shares, but their net contributions cancel.

Invariance of the full state can be checked directly in the joint exchange. In $\rho_\beta \otimes \eta_T$, the weights at the two connected endpoints satisfy

$$\frac{a_{a\nu}}{Z_K} e^{-\beta E_\nu} t_u = \frac{a_{a\nu}}{Z_K} e^{-\beta E_a} t_v. \tag{2.22}$$

The joint density operator is proportional to the identity in this two-dimensional subspace and is therefore preserved by the exchange rotation. The environment blocks are mutually orthogonal, and free evolution and return phases also preserve ρ_β. Writing $X_\beta = \sum_\nu e^{-\beta E_\nu} P_\nu / Z_K$ and $y_\beta = Z_{\mathcal{C}}/Z_K$, the remaining restoration step gives

$$\begin{aligned} \rho_\beta &= X_\beta + y_\beta \sigma_\pi^{\mathrm{sh}}, \\ \mathcal{F}_g(\rho_\beta) &= X_\beta + y_\beta \sigma_g^{\mathrm{sh}}. \end{aligned} \tag{2.23}$$

At finite temperature, $y_\beta > 0$, and the projections P_{e_a} are linearly independent. Therefore,

$$\boxed{\mathcal{F}_g(\rho_\beta) = \rho_\beta \quad \Longleftrightarrow \quad g_a = \pi_a \text{ for all } a.} \tag{2.24}$$

This is the same-temperature equilibrium condition for the selected complete cycle.

§2.4 Establishment of Equilibrium

Preservation of the equilibrium distribution by one cycle must be accompanied by convergence from other initial states. Retain the independent environment refreshing conditions of Section 1.2, fix finite β and the restoration preparation, keep $a_0 > 0$, and provide at least one positive exchange channel for each matter mode. Then every $\epsilon_\nu > 0$.

Let $k_a = 1 - \sum_\nu f_{\nu a}$ be the share that enters background restoration when starting from background mode e_a. Since $\sum_{a,\nu} a_{a\nu} = 1 - a_0$, we have $k_a \geq a_0$. The total effect associated with entry into restoration in one cycle is

$$\begin{aligned} D &= \sum_a k_a P_{e_a} + \sum_\nu \epsilon_\nu P_\nu, \\ c &= \min\{a_0, \min_\nu \epsilon_\nu\} > 0, \qquad D \geq cI. \end{aligned} \tag{2.25}$$

For any initial state, a share of at least c enters the same background preparation in each cycle. For an arbitrary input operator A, subtracting $c\,\mathrm{Tr}(A)\sigma_g^{\mathrm{sh}}$ from the restoration output leaves a completely positive map: if $D = \sum_i \delta_i |q_i\rangle\langle q_i|$, its restoration part has Kraus operators $L_{ai} = \sqrt{g_a(\delta_i - c)}|e_a\rangle\langle q_i|$. The remaining matter-return component is unchanged, and the remaining total trace is $(1-c)\,\mathrm{Tr}\,A$. Thus the common-restoration decomposition of the complete cycle [1] gives

$$\mathcal{F}_g(A) = c\,\mathrm{Tr}(A)\sigma_g^{\mathrm{sh}} + (1-c)\mathcal{G}_g(A), \tag{2.26}$$

Here $\mathcal{G}_g$ is completely positive and trace-preserving. Taking the difference of two normalized states cancels the common background term. Trace-norm contraction of positive trace-preserving maps on self-adjoint operators then gives, cycle by cycle,

$$\begin{aligned} &\|\mathcal{F}_g^n(\rho) - \mathcal{F}_g^n(\rho')\|_1 \\ &\qquad \leq (1-c)^n \|\rho - \rho'\|_1. \end{aligned} \tag{2.27}$$

This estimate shows that the complete cycle is a strictly contractive quantum channel [6] and controls both occupation and coherence deviations of the full density operator. Setting $\rho' = \rho_\beta$, whose invariance has been proved, and $g = \pi$, we obtain

$$\boxed{\begin{aligned} &\|\mathcal{F}_\pi^n(\rho_0) - \rho_\beta\|_1 \\ &\qquad \leq (1-c)^n \|\rho_0 - \rho_\beta\|_1 \longrightarrow 0. \end{aligned}} \tag{2.28}$$

Every initial state converges to the same thermal state. If another stationary state existed, substituting it into this estimate would force it to equal ρ_β; the equilibrium state is therefore unique. For general g, the same contraction also guarantees a unique stationary state: successive state differences decrease as $(1-c)^n$, their tail sums converge, and the iteration is a Cauchy sequence in the finite-dimensional state space. Thermalization additionally requires the same-temperature preparation condition of the preceding section.

Common restoration in each cycle gradually removes the system's dependence on its initial state. The joint outputs still preserve the correlations generated by the action, making local equilibration compatible with reversible joint evolution [1]. Quantum homogenization by repeated exchange also demonstrates transfer of initial-state information into joint correlations [7]. Here c gives a uniform convergence estimate, while particular energy distributions, coherences, and internal labels

can decay at different rates. The next chapter obtains these time scales from the nonstationary modes of the actual evolution.

§3 Relaxation Time Scales

Thermal equilibrium specifies the endpoint of repeated exchange, while relaxation describes how the different deviations from that endpoint decay. In one system, the total matter share, mode composition, and coherence can change at different rates. Other observables may retain information about the initial state even after the mean energy approaches equilibrium. We obtain these time scales from the complete cycle of Chapter 2 [1], using the evolution spectrum to analyze mixing times [8].

Throughout this chapter, fix the temperature and all cycle data, use same-temperature restoration $g = \pi$, choose propagation that does not mix matter modes, and retain positive exchange for each mode and $a_0 > 0$. Take $0 < \alpha_{a\nu} < \pi/2$. After one cycle, every initial state enters the closed form $\rho = X + y\sigma_\pi^{\mathrm{sh}}$; we take such a cycle endpoint as the time origin. Denote the total elapsed duration of one cycle by τ_{tot}, including exchange, return, restoration, and the required preparation operations. The nth endpoint is therefore at $t_n = n\tau_{\mathrm{tot}}$.

§3.1 Decay of Occupation Deviations

Retain $\epsilon_\nu = \sum_a \epsilon_{a\nu}$, $b_\nu = \sum_a \pi_a f_{\nu a}$, and $f_\Sigma = \sum_\nu b_\nu$. Let $N = \dim \mathcal{M}$, and arrange matter occupations and the total background share in the column vector $v = (x_1, \ldots, x_N, y)^{\mathsf{T}}$. The balance equations of Chapter 2 directly give

$$v_{n+1} = \mathsf{T} v_n, \qquad \mathsf{T} = \begin{pmatrix} I - D_\epsilon & b \\ \epsilon^{\mathsf{T}} & 1 - f_\Sigma \end{pmatrix}, \qquad D_\epsilon = \mathrm{diag}(\epsilon_1, \ldots, \epsilon_N). \tag{3.1}$$

Each column describes all possible destinations of one initial component. Matter exits with share ϵ_ν and returns with share $1 - \epsilon_\nu$. The background forms mode ν with share b_ν, while the remainder enters restoration. The equilibrium vector v_* satisfies $\mathsf{T} v_* = v_*$, so the deviation $\delta v_n = v_n - v_*$ obeys the same linear evolution.

The equilibrium relation $\epsilon_\nu x_\nu^* = b_\nu y_*$ allows these occupation equations to be symmetrized. Let $D_* = \mathrm{diag}(v_*)$ be the matrix with the equilibrium components on its diagonal. Then

$$\mathsf{S} = D_*^{-1/2} \mathsf{T} D_*^{1/2} = \begin{pmatrix} I - D_\epsilon & \boldsymbol{h} \\ \boldsymbol{h}^{\mathsf{T}} & 1 - f_\Sigma \end{pmatrix}, \qquad h_\nu = \sqrt{\epsilon_\nu b_\nu}. \tag{3.2}$$

Both exit and formation shares enter the eigenvalues. Let $\mathbf{e}_\nu$ and $\mathbf{e}_y$ denote the coordinate basis of occupation vectors, and define $\ell_\nu = \sqrt{\epsilon_\nu}\mathbf{e}_\nu - \sqrt{b_\nu}\mathbf{e}_y$. We have

$$
\begin{aligned}
I - \mathsf{S} &= \sum_{\nu} \ell_\nu \ell_\nu^{\mathsf{T}} \geq 0, \\
\mathrm{Tr}(I - \mathsf{S}) &= \sum_{\nu} \epsilon_\nu + f_\Sigma \\
&\leq \sum_{a,\nu} (\epsilon_{a\nu} + f_{\nu a}) \leq 1 - a_0.
\end{aligned}
\tag{3.3}
$$

All eigenvalues are therefore real and lie in $[a_0, 1]$. Since every mode is connected to the background, eigenvalue 1 occurs only along the normalized equilibrium direction; all other eigenvalues satisfy $a_0 \leq \lambda_r < 1$. Each occupation mode decays cycle by cycle, although individual occupations formed by superposing several modes can still vary nonmonotonically. For general parameters, all are determined by the same finite matrix, whose characteristic polynomial is

$$
\begin{aligned}
&\det(\lambda I - \mathsf{T}) \\
&= (\lambda - 1 + f_\Sigma) \prod_{\nu} (\lambda - 1 + \epsilon_\nu) \\
&\quad - \sum_{\nu} \epsilon_\nu b_\nu \prod_{\mu \neq \nu} (\lambda - 1 + \epsilon_\mu).
\end{aligned}
\tag{3.4}
$$

This follows by eliminating the diagonal matter block against the background row. Written as a polynomial, it also includes repeated roots when different exit shares coincide.

Choose orthonormal eigenvectors u_r of S, with $u_0 = \sqrt{v_*}$ corresponding to equilibrium. Normalization makes the weighted deviation $D_*^{-1/2} \delta v_0$ orthogonal to that direction. Hence,

$$
\begin{aligned}
\delta v_n &= D_*^{1/2} \sum_{r=1}^{N} A_r \lambda_r^n u_r, \\
A_r &= u_r^{\mathsf{T}} D_*^{-1/2} \delta v_0.
\end{aligned}
\tag{3.5}
$$

Each weighted vector $D_*^{1/2} u_r$ specifies a coordinated change of occupations: some matter modes increase while other modes or the background decrease, preserving total probability. This combination is multiplied by λ_r in every cycle. Expressed using the same total elapsed duration,

$$
\boxed{
\begin{gathered}
\lambda_r^n = e^{-\Gamma_r t_n}, \qquad \Gamma_r = -\frac{\log \lambda_r}{\tau_{\mathrm{tot}}}, \\
\tau_r = \Gamma_r^{-1}.
\end{gathered}
}
\tag{3.6}
$$

The quantity τ_r is the exponential time scale over which this component falls to e^{-1} of its initial amplitude. These relations describe cycle endpoints exactly. Only when $1 - \lambda_r \ll 1$ can one use $\Gamma_r \simeq (1 - \lambda_r)/\tau_{\mathrm{tot}}$. A single-exchange counting share, a per-cycle decay factor, and a relaxation rate per unit time are thus separately specified.

§3.2 Energy and Mode Composition

Mean energy probes only part of the occupation deviation. Number the matter modes as $\nu_1, \dots, \nu_N$, and let $\overline{E}_\pi = \sum_a \pi_a E_a$ be the mean background energy. Arrange the component energies in the column vector $e = (E_{\nu_1}, \dots, E_{\nu_N}, \overline{E}_\pi)^{\mathsf{T}}$. Substitution into the preceding expansion yields

$$\begin{aligned} E_S(n) - E_S^* &= e^{\mathsf{T}} \delta v_n \\ &= \sum_{r=1}^{N} A_r \big(e^{\mathsf{T}} D_*^{1/2} u_r\big) \lambda_r^n. \end{aligned} \tag{3.7}$$

For a slow mode to appear in the energy curve, it must be excited by the initial state and have a nonzero energy response. When occupations are redistributed among modes of equal energy, the mean energy can remain unchanged even though their detection probabilities continue to vary. Verifying complete thermalization therefore also requires examining these internal occupations.

This distinction can be solved exactly when $N \geq 2$ and all matter modes have the same exit share $\epsilon_\nu = \epsilon > 0$. The background equation reduces to

$$\begin{aligned} y_{n+1} &= \epsilon + (1 - \epsilon - f_\Sigma) y_n, \\ y_* &= \frac{\epsilon}{\epsilon + f_\Sigma}, \\ y_n - y_* &= (1 - \epsilon - f_\Sigma)^n (y_0 - y_*). \end{aligned} \tag{3.8}$$

The total matter share $1 - y_n$ approaches equilibrium at the same rate. Now let $r_\nu = b_\nu / f_\Sigma$, and define deviations of the internal matter occupations by

$$\begin{aligned} z_\nu(n) &= x_\nu(n) - r_\nu(1 - y_n), \\ \sum_\nu z_\nu(n) &= 0. \end{aligned} \tag{3.9}$$

Substitute $x_\nu(n+1) = (1 - \epsilon) x_\nu(n) + b_\nu y_n$ and the background equation. Using $r_\nu f_\Sigma = b_\nu$, all formation terms cancel, leaving

$$\boxed{\begin{aligned} z_\nu(n) &= (1 - \epsilon)^n z_\nu(0), \\ x_\nu(n) &= r_\nu(1 - y_n) + (1 - \epsilon)^n z_\nu(0). \end{aligned}} \tag{3.10}$$

The total share and internal composition thus have two different decay factors:

$$\begin{aligned} \lambda_{\text{tot}} &= 1 - \epsilon - f_\Sigma, & \Gamma_{\text{tot}} &= -\frac{\log \lambda_{\text{tot}}}{\tau_{\text{tot}}}, \\ \lambda_{\text{comp}} &= 1 - \epsilon, & \Gamma_{\text{comp}} &= -\frac{\log \lambda_{\text{comp}}}{\tau_{\text{tot}}}. \end{aligned} \tag{3.11}$$

The internal composition has $N - 1$ independent components. Since $f_\Sigma > 0$, we have $\Gamma_{\text{tot}} > \Gamma_{\text{comp}}$: the total matter and background shares balance more rapidly, while the relative occupations of matter modes can continue to adjust. This rate difference is already contained in the same exit and formation shares.

Writing $E_{\mathcal{M}} = \sum_\nu r_\nu E_\nu$, the corresponding energy curve is

$$E_S(n) - E_S^* = (\overline{E}_\pi - E_{\mathcal{M}})(y_0 - y_*)\lambda_{\text{tot}}^n + \lambda_{\text{comp}}^n \sum_\nu E_\nu z_\nu(0). \tag{3.12}$$

If these matter modes have equal energies, the second term vanishes because $\sum_\nu z_\nu = 0$. The energy curve then shows only the faster relaxation of the total share, while measurements of internal modes still reveal the slower composition change.

§3.3 Coherence Decay

Coherences between matter modes evolve through the same cycle. Thermal exchange changes each mode's return amplitude, and the full environment output retains differences between these amplitudes. Tracing out the environment then changes the off-diagonal entries. Background restoration produces only the shell-average component, and renewed formation adds only diagonal matter occupations. Neither supplies coherence between different matter modes [1].

Let $D_{a\lambda} = I_{\mathcal{M}} + (\cos\alpha_{a\lambda} - 1)P_\lambda$ be the diagonal return operator for channel (a, λ). For environment input u, it gives the matter-return amplitude; for input v, the existing matter component is preserved. Denote the net cycle phase from free evolution and coherent retrieval by φ_ν. Its action on the matter space is $U_\varphi = \sum_\nu e^{i\varphi_\nu} P_\nu$. With this phase removed, the complete return map is

$$\begin{aligned} &U_\varphi^\dagger \mathcal{Q}(X) U_\varphi \\ &= a_0 X + \sum_{a,\lambda} a_{a\lambda} \left[t_u D_{a\lambda} X D_{a\lambda} + t_v X \right]. \end{aligned} \tag{3.13}$$

For $\nu \neq \mu$, a channel changes at most one side of $|\nu\rangle\langle\mu|$. Summing channel contributions and defining

$$\begin{aligned} d_\nu &= \sum_a a_{a\nu} t_u (1 - \cos\alpha_{a\nu}), \\ \gamma_{\nu\mu} &= 1 - d_\nu - d_\mu, \end{aligned} \tag{3.14}$$

gives the exact coherence factor of the complete cycle and its iteration:

$$\begin{aligned} \mathcal{F}(|\nu\rangle\langle\mu|) &= \gamma_{\nu\mu} e^{i(\varphi_\nu - \varphi_\mu)} |\nu\rangle\langle\mu|, \\ \rho_{\nu\mu}(n) &= \gamma_{\nu\mu}^n e^{in(\varphi_\nu - \varphi_\mu)} \rho_{\nu\mu}(0). \end{aligned} \tag{3.15}$$

The chosen exchange angles and positive channel weights ensure $a_0 \leq \gamma_{\nu\mu} < 1$. The magnitude of the off-diagonal entry decays at the rate

$$\boxed{\Gamma_{\nu\mu}^{\text{coh}} = -\frac{\log \gamma_{\nu\mu}}{\tau_{\text{tot}}}} \tag{3.16}$$

while the difference in net cycle phases makes interference readings oscillate from one cycle to the next. For example, the expectation of $A_{\nu\mu} = |\mu\rangle\langle\nu| + |\nu\rangle\langle\mu|$ is

$$\langle A_{\nu\mu} \rangle_n = 2\gamma_{\nu\mu}^n \, \text{Re}\left[e^{in(\varphi_\nu - \varphi_\mu)} \rho_{\nu\mu}(0) \right]. \tag{3.17}$$

Coherence decay therefore appears as a change in the envelope of an actual interference signal. This signal and the energy curve read off-diagonal and diagonal components of the same full state. The different constraints on coherence and occupations in thermal processes can also be analyzed through time-translation symmetry [9].

Exchange shares and return amplitudes yield a direct comparison. Applying $\sin^2\alpha = (1-\cos\alpha)(1+\cos\alpha)$ channel by channel gives

$$d_\nu = \sum_a \frac{\epsilon_{a\nu}}{1+\cos\alpha_{a\nu}}, \qquad \frac{\epsilon_\nu}{2} < d_\nu < \epsilon_\nu. \tag{3.18}$$

For the equal exit shares of the preceding section, $\gamma_{\nu\mu} < 1-\epsilon$, so coherence decays faster than the internal composition. Its ordering relative to total-share relaxation still depends on the comparison between $d_\nu + d_\mu$ and $\epsilon + f_\Sigma$. For unequal exit shares, one must calculate the occupation spectrum and coherence factors separately before comparing the time scales of actual readings.

§3.4 Continuous Relaxation under Weak Exchange

When exchange per cycle is sufficiently weak, many complete cycles accumulate into continuous changes in occupations and coherences. Continuous limits of repeated interactions and their weak-coupling time scales have been studied rigorously [10,11]. Keep all other cycle data fixed and set only the exchange angles to $\alpha_{a\nu}(h) = \sqrt{h}\,\theta_{a\nu}$, where $h > 0$ is dimensionless and $\theta_{a\nu}$ is fixed. Starting from a fixed state $X_0 + y_0\sigma_\pi^{\mathrm{sh}}$ in the closed family, remove the accumulated diagonal phase at the endpoints: $\widetilde{X}_n = U_\varphi^{-n} X_n U_\varphi^n$. Return, exit, and diagonal formation are compatible with this transformation, and tildes are omitted below. Introduce slow time and two sets of rates:

$$\begin{aligned} s_n &= h t_n = h n \tau_{\mathrm{tot}}, \\ \kappa_\nu^{\mathrm{out}} &= \frac{1}{\tau_{\mathrm{tot}}} \sum_a a_{a\nu} t_u^{(a\nu)} \theta_{a\nu}^2, \\ \kappa_\nu^{\mathrm{in}} &= \frac{1}{\tau_{\mathrm{tot}}} \sum_a \pi_a a_{a\nu} t_v^{(a\nu)} \theta_{a\nu}^2. \end{aligned} \tag{3.19}$$

The expansions $\sin^2(\sqrt{h}\,\theta) = h\theta^2 + O(h^2)$ and $1-\cos(\sqrt{h}\,\theta) = h\theta^2/2 + O(h^2)$ give the per-cycle shares and coherence losses as

$$\begin{aligned} \epsilon_\nu(h) &= h\tau_{\mathrm{tot}}\kappa_\nu^{\mathrm{out}} + O(h^2), \\ b_\nu(h) &= h\tau_{\mathrm{tot}}\kappa_\nu^{\mathrm{in}} + O(h^2), \\ d_\nu(h) &= \tfrac{1}{2} h\tau_{\mathrm{tot}}\kappa_\nu^{\mathrm{out}} + O(h^2). \end{aligned} \tag{3.20}$$

The finite number of fixed channels makes the remainders uniformly controlled. Write $\mathsf{K}_{\mathrm{out}} = \sum_\nu \kappa_\nu^{\mathrm{out}} P_\nu$, $\mathsf{K}_{\mathrm{in}} = \sum_\nu \kappa_\nu^{\mathrm{in}} P_\nu$, and $\kappa_\Sigma^{\mathrm{in}} = \sum_\nu \kappa_\nu^{\mathrm{in}}$. Substituting into the exact occupation and off-diagonal recursions and dividing successive endpoint differences by $h\tau_{\mathrm{tot}}$ gives the slow-time equations

$$\boxed{\begin{aligned}\frac{dX}{ds} &= -\tfrac{1}{2}\{\mathsf{K}_{\mathrm{out}}, X\} + y\mathsf{K}_{\mathrm{in}},\\ \frac{dy}{ds} &= \mathrm{Tr}(\mathsf{K}_{\mathrm{out}}X) - \kappa_{\Sigma}^{\mathrm{in}} y.\end{aligned}} \tag{3.21}$$

Exit reduces the existing matter component, and reverse formation adds diagonal occupations. Their contributions to the total background share have the opposite signs. Off-diagonal entries satisfy $dX_{\nu\mu}/ds = -(\kappa_\nu^{\mathrm{out}} + \kappa_\mu^{\mathrm{out}})X_{\nu\mu}/2$, so their decay is determined by the same exit rates.

These equations have an explicit completely positive evolution representation. Use a mathematical label $|\star\rangle$ to encode the total background share, write $\widehat{\rho} = X \oplus y$ on $\widehat{\mathcal{H}} = \mathcal{M} \oplus \mathbb{C}|\star\rangle$, and set

$$\begin{aligned} J_\nu^{\mathrm{out}} &= \sqrt{\kappa_\nu^{\mathrm{out}}}|\star\rangle\langle\nu|,\\ J_\nu^{\mathrm{in}} &= \sqrt{\kappa_\nu^{\mathrm{in}}}|\nu\rangle\langle\star|,\\ \mathcal{D}[J](A) &= JAJ^\dagger - \tfrac{1}{2}\{J^\dagger J, A\},\\ \mathcal{L} &= \sum_\nu \big(\mathcal{D}[J_\nu^{\mathrm{out}}] + \mathcal{D}[J_\nu^{\mathrm{in}}]\big). \end{aligned} \tag{3.22}$$

This Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) generator [12,13] gives exactly the preceding equations on block-diagonal states $\widehat{\rho}$. The label $|\star\rangle$ encodes only the total share. The physical background remains the distribution σ_π^{sh}, rather than becoming a single energy level. The same-temperature relation gives, channel by channel,

$$\frac{\kappa_\nu^{\mathrm{in}}}{\kappa_\nu^{\mathrm{out}}} = \frac{e^{-\beta E_\nu}}{Z_{\mathcal{C}}}, \qquad x_\nu^* = \frac{e^{-\beta E_\nu}}{Z_K}, \quad y_* = \frac{Z_{\mathcal{C}}}{Z_K}. \tag{3.23}$$

If every mode has at least one channel with $\theta_{a\nu} > 0$, exit rates are positive, the occupation chain is connected, and coherences decay. The limiting equations therefore still converge to the same full thermal state.

Let Φ_h be the single-cycle map after phase removal and encoding, restricted to the above block-diagonal operator space. Expanding the finite-dimensional cycle map and the semigroup exponential gives $\|\Phi_h - e^{h\tau_{\mathrm{tot}}\mathcal{L}}\|_{1\to1,\mathrm{sa}} \le C_0 h^2$, where the norm acts on self-adjoint operators in that space. Both evolutions preserve this space and contract the trace norm on its self-adjoint operators. Subtracting successive iterates and using a telescoping sum yields

$$\begin{aligned} &\|\Phi_h^n(\widehat{\rho}_0) - e^{s_n\mathcal{L}}\widehat{\rho}_0\|_1\\ &\qquad \le nC_0h^2 \le \frac{C_0 S}{\tau_{\mathrm{tot}}}h, \qquad 0 \le s_n \le S. \end{aligned} \tag{3.24}$$

This proves uniform convergence on every fixed finite slow-time interval after removal of the net cycle phase. Taking $n = \lfloor s/(h\tau_{\mathrm{tot}})\rfloor$ leaves less than one cycle of slow time, adding only an $O(h)$ error. Reconstructing both states as $X + y\sigma_\pi^{\mathrm{sh}}$ and restoring the accumulated phase U_φ^n to both matter blocks preserve the trace norm of their difference. The same error bound therefore holds when comparing the original physical state with the corresponding rotating reference state.

Weak exchange does not shorten the total elapsed duration of return and restoration. Let $P_{\mathcal{M}}$ and $P_{\mathcal{C}}$ project onto the two subspaces, extend the net cycle phase

action as $\widehat{U}_\varphi = P_{\mathcal{C}} + U_\varphi$, and write $\mathcal{U}_\varphi(A) = \widehat{U}_\varphi A \widehat{U}_\varphi^\dagger$. The single-cycle limit on the full space is then

$$\begin{aligned} \mathcal{F}_h &\longrightarrow \mathcal{U}_\varphi \circ \mathcal{E}_\pi, \\ \mathcal{E}_\pi(A) &= P_{\mathcal{M}} A P_{\mathcal{M}} + \mathrm{Tr}(P_{\mathcal{C}} A)\sigma_\pi^{\mathrm{sh}}. \end{aligned} \tag{3.25}$$

Background restoration remains after phase removal. This limit is the identity only on the restored closed family of states. The continuous equations thus describe slow changes accumulated over many weak-exchange cycles. The coherence decay rate in the original physical time is correspondingly

$$\Gamma_{\nu\mu}^{\mathrm{coh}}(h) = h\,\frac{\kappa_\nu^{\mathrm{out}} + \kappa_\mu^{\mathrm{out}}}{2} + O(h^2). \tag{3.26}$$

§3.5 Slow Modes and Thermalization Time

Return to the complete cycle with fixed exchange angles and physical time $t_n = n\tau_{\mathrm{tot}}$. After the first restoration, state deviations consist of occupation and coherence modes. Long-time readings are controlled by the slowest component that is both excited by the initial state and accessible to the observable.

Including all modes, define

$$\begin{aligned} r_{\mathrm{slow}} &= \max\left\{ \max_{r\geq 1} \lambda_r, \max_{\nu\neq\mu} \gamma_{\nu\mu} \right\} < 1, \\ \boxed{\Gamma_{\mathrm{slow}} &= -\frac{\log r_{\mathrm{slow}}}{\tau_{\mathrm{tot}}}.} \end{aligned} \tag{3.27}$$

If there is only one matter mode, the coherence term is absent. The common-restoration lower bound in Chapter 2 also ensures

$$r_{\mathrm{slow}} \leq 1 - c, \qquad \Gamma_{\mathrm{slow}} \geq -\frac{\log(1-c)}{\tau_{\mathrm{tot}}}. \tag{3.28}$$

This uniform estimate guarantees decay of deviations, while the actual thermalization rate is determined by the calculated spectrum. If the late-time envelope of a reading is dominated by $C_A e^{-\Gamma_A t_n}$, the time needed to reach error δ is approximately

$$t_\delta \simeq \frac{1}{\Gamma_A} \log \frac{|C_A|}{\delta}, \tag{3.29}$$

The actual endpoint is the first complete-cycle count no earlier than this time. Greater precision resolves slow components previously obscured by faster processes.

When mode rates are well separated, energy readings may already have stabilized while the internal composition still retains information about the initial preparation. Spectral separation of time scales also underlies metastability in open quantum dynamics [14]. Let Γ_{f} and Γ_{s} be the envelope decay rates of selected fast and slow modes. In units of their respective initial amplitudes, requiring the fast mode to fall below δ and the slow mode to change by no more than δ, with $0 < \delta < 1$, gives $e^{-\Gamma_{\mathrm{f}} t_n} \leq \delta$ and $1 - e^{-\Gamma_{\mathrm{s}} t_n} \leq \delta$. Both conditions hold in the time window

$$\boxed{\frac{\log(1/\delta)}{\Gamma_{\mathrm{f}}} \leq t_n \leq \frac{-\log(1-\delta)}{\Gamma_{\mathrm{s}}}.} \tag{3.30}$$

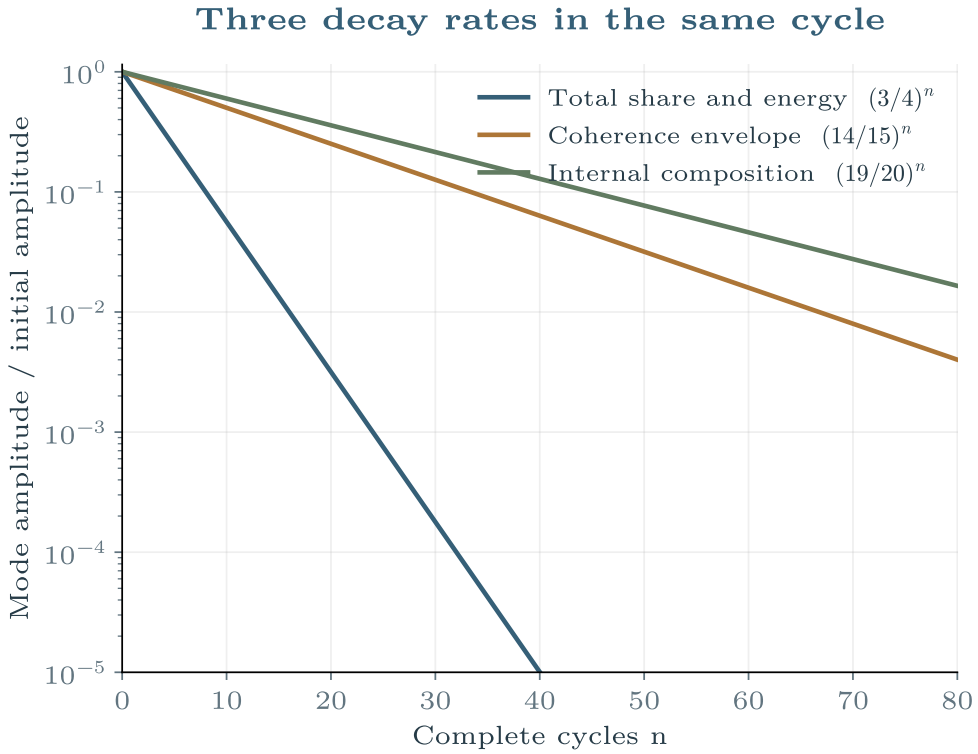


Figure 3-1. **Different relaxation rates in the same cycle.** The example has five equal-energy matter modes and satisfies the channel-weight constraint. Each mode exchanges only with one equal-energy background endpoint, with $a_{a\nu} = 2/15$, $t_u = t_v = 1/2$, and $\alpha_{a\nu} = \pi/3$. Thus $a_0 = 1/3$ and $\epsilon = 1/20$. A second background endpoint lies one shell higher, with energy difference $\log p$. Taking $\beta \log p = \log 4$ gives same-temperature restoration shares $(4/5, 1/5)$ and $f_\Sigma = 1/5$. The curves $(3/4)^n$, $(14/15)^n$, and $(19/20)^n$ show mode amplitudes relative to their initial values on a logarithmic vertical scale. They compare decay rates for this specified cycle; the actual initial amplitudes of the different mode types are set by the preparation.

This interval must contain an actual cycle endpoint. Readings that superpose several modes also require the actual amplitudes of all excited modes to be included.

How long a slow component persists also depends on the energy exchange needed to change it. If exit requires absorption of energy from the environment, cooling reduces the opportunities for such exchange. Again consider the case $N \geq 2$ with the same exit share for all matter modes. Let each matter mode exchange only with a background endpoint one shell higher, so the established shell spectrum gives $\Delta = E_a - E_\nu = \log p$. Assign the same weight a_{ex} and exchange angle α to every channel, with $N a_{\mathrm{ex}} < 1$. Then

$$\epsilon = \frac{a_{\mathrm{ex}} \sin^2 \alpha}{1 + e^{\beta\Delta}}, \qquad \Gamma_{\mathrm{comp}} = -\frac{\log(1-\epsilon)}{\tau_{\mathrm{tot}}}. \tag{3.31}$$

For $\beta\Delta \gg 1$, with weights, exchange angles, and total elapsed duration fixed,

$$\boxed{\Gamma_{\mathrm{comp}} \sim \frac{a_{\mathrm{ex}} \sin^2 \alpha}{\tau_{\mathrm{tot}}} e^{-\beta\Delta}.} \tag{3.32}$$

The activation energy is the energy Δ that the exit channel actually needs to absorb. Additional exit channels that require no energy increase can remove this exponential slowing. When several channels coexist, all must be included in the occupation matrix. The same structure can therefore ultimately thermalize at finite temperature while the relative matter-mode occupations retain information about the initial preparation for long but finite observation times.

These time scales assume a fixed environment temperature. When that temperature changes, the target thermal state moves as well. Whether the system follows it depends on the comparison between target motion and actual relaxation. This comparison connects fixed-temperature thermalization to cooling.

§4 Cooling and Temperature Evolution

As matter releases energy to the environment, its energy distribution changes and the corresponding energy-based temperature falls. The relaxation times obtained in Chapter 3 determine whether the distribution can adjust sufficiently quickly. If the target thermal state specified by the environment preparation moves slowly enough, the system can remain close to a sequence of changing thermal states. If the target moves too fast, the actual distribution lags behind it. Cooling therefore involves both energy balance and internal relaxation.

The energy operator H_K remains fixed in this chapter. Denote the inverse temperature of the environment preparation by β_{B}, and the inverse temperature obtained by inverting the system's mean-energy relation by β_E. The former enters the exchange and restoration preparations; the latter is read from the system state.

§4.1 Temperature and Energy Distribution

On the full state space, the equilibrium energy at inverse temperature β is

$$
\begin{aligned}
Z_K(\beta) &= \sum_{j=0}^{K} d_j e^{-\beta E_j}, \qquad d_j = |S_j|, \\
U_K(\beta) &= \mathrm{Tr}(H_K \rho_\beta) = -\partial_\beta \log Z_K(\beta).
\end{aligned}
\tag{4.1}
$$

All independent states at each energy are included. Differentiating U_K once more gives

$$
\boxed{
\begin{aligned}
\frac{dU_K}{d\beta} &= -V_K(\beta), \\
V_K(\beta) &= \mathrm{Tr}(H_K^2 \rho_\beta) - U_K(\beta)^2.
\end{aligned}
}
\tag{4.2}
$$

As long as distinct energies participate at finite temperature, $V_K > 0$, and the equilibrium energy decreases strictly with β. In terms of $T = 1/\beta$, the heat capacity is

$$
C_K(T) = \frac{dU_K}{dT} = \beta^2 V_K(\beta) > 0. \tag{4.3}
$$

The actual state ρ_n has mean energy $E_n = \mathrm{Tr}(H_K \rho_n)$. On the positive-temperature branch $0 < E_n < U_K(0)$, the equations

$$
U_K(\beta_E(n)) = E_n, \qquad T_E(n) = \frac{1}{\beta_E(n)} \tag{4.4}
$$

define a unique energy-based temperature. This provides a common reading for cooling curves. Whether the full distribution is close to the corresponding Gibbs state is still determined by $\rho_n - \rho_{\beta_E(n)}$. The redistribution among equal-energy modes in Chapter 3 already shows that the same mean energy can correspond to different internal occupations.

The finite spectrum restricts the energy to $0 \le E_n \le K \log p$. The partition function gives $U_K(0) = \mathrm{Tr}\, H_K / \dim \mathcal{H}_K$ and $U_K(\beta) \to 0$ as $\beta \to +\infty$. All temperature changes below are calculated along this finite-space energy curve.

§4.2 Thermalization at Changing Temperature

Under the environment refreshing conditions of Section 1.2, let the incoming thermal environment in cycle n have inverse temperature $\beta_{\mathrm{B},n}$, and take the restoration shares at the same temperature, $g_n = \pi(\beta_{\mathrm{B},n})$. Fix the other channel data and the total elapsed duration. Denote the actual cycle and its equilibrium state by

$$
\begin{aligned}
\mathcal{F}_n &= \mathcal{F}_{\beta_{\mathrm{B},n}}, \qquad & \rho_n^* &= \rho_{\beta_{\mathrm{B},n}}, \\
\rho_{n+1} &= \mathcal{F}_n(\rho_n), \qquad & \mathcal{F}_n(\rho_n^*) &= \rho_n^*.
\end{aligned}
\tag{4.5}
$$

Each cycle preserves its own target thermal state, but the next target can differ. Tracking instantaneous stationary states under slowly varying parameters is also studied in the adiabatic theory of contracting evolutions [15]. Setting $\delta_n = \rho_n - \rho_n^*$ and subtracting directly gives

$$
\boxed{\delta_{n+1} = \mathcal{F}_n(\delta_n) + \rho_n^* - \rho_{n+1}^*.} \tag{4.6}
$$

The first term is the decay of the existing deviation during the current cycle; the second is the new deviation caused by target motion. Together they determine the actual lag.

Using the contraction decomposition of Chapter 2, let $c_n > 0$ be the common-restoration lower bound for this cycle. With $e_n = \|\delta_n\|_1$ and $d_n = \|\rho^*_{n+1} - \rho^*_n\|_1$, we obtain

$$\begin{aligned} e_{n+1} &\le (1 - c_n)e_n + d_n, \\ e_n &\le e_0 \prod_{k=0}^{n-1}(1 - c_k) \\ &\quad + \sum_{m=0}^{n-1} d_m \prod_{k=m+1}^{n-1} (1 - c_k). \end{aligned} \tag{4.7}$$

An empty product equals one. Each deviation produced by target motion is attenuated by the subsequent cycles.

Since H_K is fixed, the Gibbs-state derivative can be calculated separately at each energy:

$$\begin{aligned} \partial_\beta \rho_\beta &= -(H_K - U_K(\beta))\rho_\beta, \\ \|\partial_\beta \rho_\beta\|_1 &= \mathrm{Tr}(|H_K - U_K(\beta)|\rho_\beta) \\ &\le \sqrt{V_K(\beta)}. \end{aligned} \tag{4.8}$$

Thus, if I_n is the interval between the inverse temperatures of adjacent cycles, the target displacement satisfies

$$d_n \le |\beta_{\mathrm{B},n+1} - \beta_{\mathrm{B},n}| \sup_{\beta \in I_n} \sqrt{V_K(\beta)}. \tag{4.9}$$

On a fixed finite-temperature interval, if the actual channels maintain $c_n \ge c_{\min} > 0$ and $d_n \le d_{\max}$, the exact accumulated bound reduces to

$$\boxed{\begin{aligned} e_n &\le (1 - c_{\min})^n e_0 \\ &\quad + \frac{d_{\max}}{c_{\min}}\big[1 - (1 - c_{\min})^n\big]. \end{aligned}} \tag{4.10}$$

Once the initial-state contribution has decayed, $d_{\max}/c_{\min} \ll 1$ is sufficient for tracking. When the temperature varies smoothly with sufficiently small changes per cycle, the corresponding approximate criterion is that the following ratio remain uniformly small over the interval considered:

$$\frac{\tau_{\mathrm{tot}} |\dot{\beta}_{\mathrm{B}}| \sqrt{V_K(\beta_{\mathrm{B}})}}{c_{\min}} \ll 1. \tag{4.11}$$

This sufficient condition controls the full-state deviation. If exit channels weaken upon cooling, tracking should instead be checked using the cycle-dependent c_n and the accumulated bound.

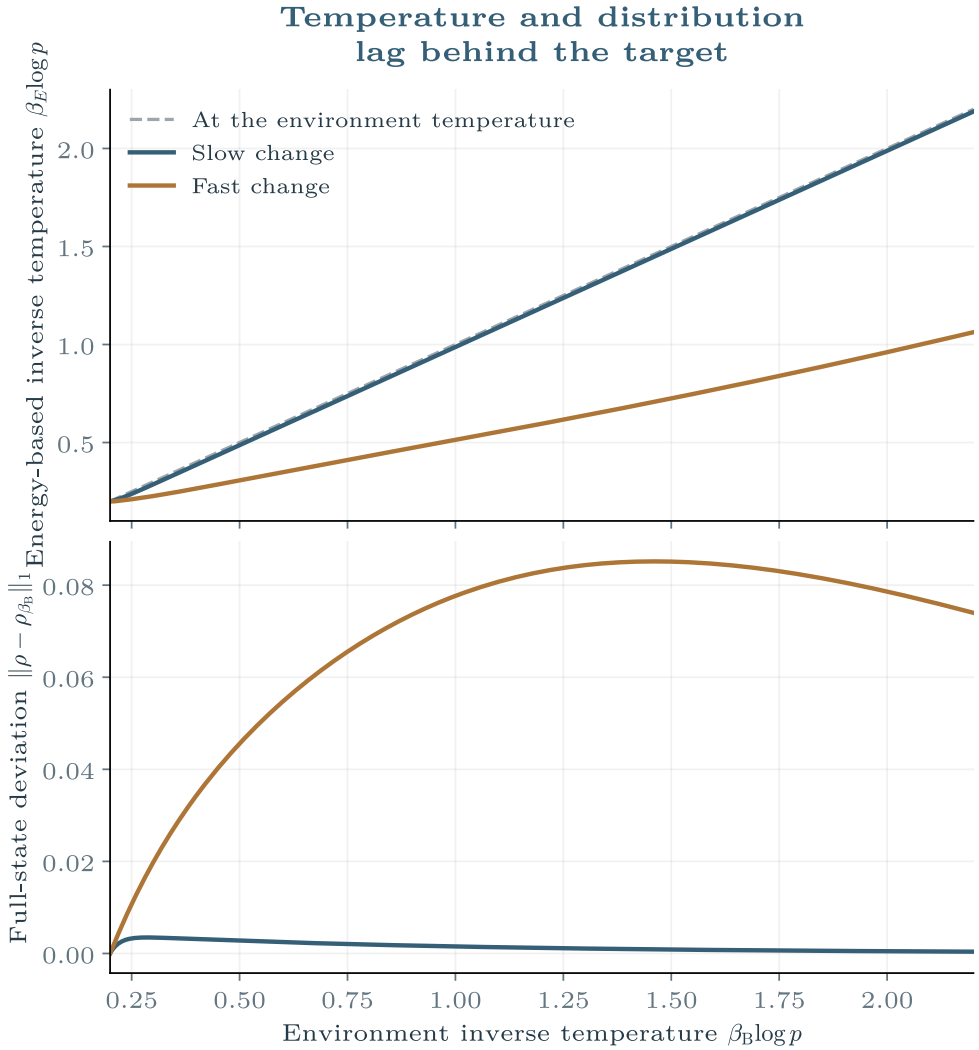


Figure 4-1. **Temperature change and distribution lag.** The environment inverse temperature is increased at two rates in the same finite cycle. The upper panel compares the environment parameter $\beta_B \log p$ with the energy-based inverse-temperature reading $\beta_E \log p$; the lower panel gives the trace-norm distance between the full diagonal state and the target Gibbs state for that cycle. Both processes start from the same thermal state. Slower change allows more cycles for the distribution to adjust. The example uses a 49-dimensional state space at truncation depth $K = 2$. Every matter mode exchanges with the zero-energy background endpoint, with channel weight 0.015 and exchange angle 0.8. The environment steps per cycle are $\Delta(\beta_B \log p) = 10^{-4}$ and 10^{-2}, respectively. The calculation always retains the actual background distribution from the preceding cycle.

§4.3 Energy Flow and Cooling Rate

Changes in environment temperature affect the system through actual exchanges. At the beginning of cycle n, denote the matter occupations by $x_\nu(n)$ and the background endpoint occupations by $z_a(n)$. When temperature changes, the incoming background still comes from the preceding output and generally differs from the restoration shares prescribed for the current cycle. The net channel exit must therefore be written as

$$J_{a\nu}(n) = \epsilon_{a\nu}(\beta_{\mathrm{B},n})x_\nu(n) - f_{\nu a}(\beta_{\mathrm{B},n})z_a(n). \tag{4.12}$$

After thermal exchange, the background occupations and their total share are

$$\widetilde{z}_a = z_a + \sum_\nu J_{a\nu}, \qquad \widetilde{y} = \sum_a \widetilde{z}_a. \tag{4.13}$$

Restoration then redistributes them as $\widetilde{y}\pi_a(\beta_{\mathrm{B},n})$. The energy balance of Chapter 2 holds cycle by cycle:

$$\begin{aligned} Q_T(n) &= \sum_{a,\nu}(E_a - E_\nu)J_{a\nu}(n), \\ Q_R(n) &= \widetilde{y}\,\overline{E}_{\pi_n} - \sum_a E_a\widetilde{z}_a, \\ \overline{E}_{\pi_n} &= \sum_a \pi_a(\beta_{\mathrm{B},n})E_a, \\ E_{n+1} - E_n &= Q_T(n) + Q_R(n). \end{aligned} \tag{4.14}$$

The mean outward energy flow must therefore be read from the complete balance:

$$P_{\mathrm{out}}(n) = -\frac{Q_T(n) + Q_R(n)}{\tau_{\mathrm{tot}}}. \tag{4.15}$$

A positive $P_{\mathrm{out}} > 0$ means that the system loses energy during this cycle, lowering its energy-based temperature. Thermal exchange and background restoration can each absorb or release energy; their sum determines the actual cooling rate. Resources needed to change the environment preparation belong to the preparation apparatus. The quantity calculated here is the energy gained or lost by the system when that preparation interacts with it.

Since $E_n = U_K(\beta_E(n))$, the temperature change between adjacent endpoints satisfies the exact relation

$$\boxed{\int_{\beta_E(n)}^{\beta_E(n+1)} V_K(b)\,db = \tau_{\mathrm{tot}}P_{\mathrm{out}}(n).} \tag{4.16}$$

Positive outward energy flow raises the upper integration limit: β_E increases and T_E decreases. On time scales for which temperature changes sufficiently little per cycle, the endpoint differences further give

$$\boxed{\begin{aligned} V_K(\beta_E)\,\dot{\beta}_E &= P_{\mathrm{out}}, \\ C_K(T_E)\,\dot{T}_E &= -P_{\mathrm{out}}. \end{aligned}} \tag{4.17}$$

These relations convert actual energy output into temperature change. Energy flow follows from joint exchange and restoration, while heat capacity follows from the full Gibbs partition function. If the distribution also satisfies the tracking condition of the preceding section, the energy reading can describe the temperature of a system close to a thermal state.

§4.4 Low-Temperature Cooling

These energy relations also apply to cooling that does not maintain thermal equilibrium. Consider a cycle in which matter modes exit to the zero-energy background and release energy to the environment. Take $K \geq 2$, allow each matter mode to exchange only with the zero-energy background endpoint e_0, and take both the thermal environment and restoration preparations to the zero-temperature limit. Channel weights and exchange angles remain fixed, with $a_0 > 0$. For a positive-energy matter mode, $E_\nu > 0$ and $E_0 - E_\nu = -E_\nu$, so

$$\begin{aligned} &t_u \longrightarrow 1, \qquad t_v \longrightarrow 0, \\ &\epsilon_\nu = a_{0\nu} \sin^2 \alpha_{0\nu} > 0, \qquad f_{\nu 0} = 0. \end{aligned} \tag{4.18}$$

At zero temperature, positive-energy modes can still exit by releasing energy, while reverse formation stops because the environment supplies no energy. In contrast, the energy-absorbing exit channels at the end of Chapter 3 are suppressed upon cooling. Zero-energy matter modes still exchange with e_0 at equal energy, with $t_u = t_v = 1/2$, without affecting total system energy. The restoration preparation tends to $\sigma^{\mathrm{sh}} = P_{e_0}$, transferring all positive background energy to the restoration auxiliary.

Take the state after the first completed cooling cycle as the time origin for this section. Thereafter, only matter modes carry positive energy. They receive no reverse-formation contribution, so the full occupation equations reduce directly to

$$\begin{aligned} x_\nu(n) &= (1 - \epsilon_\nu)^n x_\nu(0) \quad (E_\nu > 0), \\ E_n &= \sum_{E_\nu > 0} E_\nu x_\nu(0)(1 - \epsilon_\nu)^n. \end{aligned} \tag{4.19}$$

If the excitation energy is initially nonzero at this time origin, it decreases through the output of each cycle and remains positive for any finite number of cycles. Let λ_E be the largest $1 - \epsilon_\nu$ among initially occupied positive-energy modes, and let $A_E > 0$ be the sum of all energy coefficients with that factor. The finite number of modes gives

$$\begin{aligned} E_n &= A_E e^{-\Gamma_E t_n}[1 + o(1)], \\ \Gamma_E &= -\frac{\log \lambda_E}{\tau_{\mathrm{tot}}}. \end{aligned} \tag{4.20}$$

This energy decay rate depends on the actual exit channels and initial state. If all positive-energy modes have the same exit share ϵ, then $E_n = E_{\mathrm{init}}(1 - \epsilon)^n$ holds from the time origin, without waiting for late-time dominance of a leading mode.

Converting this energy curve to a temperature reading requires the low-temperature form of the full spectrum. Let d_0 and d_1 be the degeneracies of the ground and first excited levels, respectively, and let $\Delta = \log p$ be the first excitation gap. For finite K,

$$
\begin{aligned}
Z_K(\beta) &= d_0 + d_1 e^{-\beta\Delta} + O(e^{-2\beta\Delta}), \\
U_K(\beta) &= \Delta \frac{d_1}{d_0} e^{-\beta\Delta}[1 + O(e^{-\beta\Delta})].
\end{aligned}
\tag{4.21}
$$

Once the actual energy enters the positive-temperature branch, inversion of $U_K(\beta_E(n)) = E_n$ gives the late-time relations

$$
\beta_E(n) = \frac{\Gamma_E}{\Delta} t_n + \frac{1}{\Delta} \log \frac{\Delta d_1}{d_0 A_E} + o(1). \tag{4.22}
$$

$$
\boxed{T_E(n) \sim \frac{\Delta}{\Gamma_E t_n}.} \tag{4.23}
$$

Exponential energy decay corresponds to an inverse-time decrease of the temperature reading. The quantity Γ_E comes from the actual slow energy mode, while Δ is the first excitation gap of the full thermal distribution. The slowest energy mode need not lie at the lowest excited level; the two enter the result separately.

The inverse-time law in this section is determined jointly by the actual zero-temperature exit rates and the finite spectrum. Changing the incoming temperature, reverse-formation channels, coupling, or duration requires the energy flow to be recalculated. Preparation time and resources needed to realize the cold source must also be included in an apparatus-level analysis; such constraints are discussed in resource-based studies of the third law of thermodynamics [16].

§4.5 Corrections to the Cooling Curve

The approach to the inverse-time law also reflects the initial preparation and the other energy modes. Let $u = e^{-\beta\Delta}$, $r_1 = d_1/d_0$, and $r_2 = d_2/d_0$. For fixed finite $K \geq 2$, expanding the full partition function to second order gives

$$
\begin{aligned}
\frac{Z_K}{d_0} &= 1 + r_1 u + r_2 u^2 + O(u^3), \\
\frac{U_K}{\Delta} &= \frac{r_1 u + 2r_2 u^2 + O(u^3)}{1 + r_1 u + r_2 u^2 + O(u^3)} \\
&= r_1 u + (2r_2 - r_1^2) u^2 + O(u^3).
\end{aligned}
\tag{4.24}
$$

Writing $x = E/\Delta$ and inverting order by order yields

$$
u = \frac{x}{r_1} - \frac{2r_2 - r_1^2}{r_1^3} x^2 + O(x^3). \tag{4.25}
$$

Using $\Delta\beta_E = -\log u$, we then obtain

$$
\boxed{
\begin{aligned}
\Delta\beta_E &= \log \frac{\Delta r_1}{E} + c_{\mathrm{G}} \frac{E}{\Delta} + O\left(\frac{E^2}{\Delta^2}\right), \\
c_{\mathrm{G}} &= \frac{2r_2}{r_1^2} - 1.
\end{aligned}
}
\tag{4.26}
$$

This coefficient is determined by the full shell degeneracies. The first excited shell has $r_1 = 1/p$. For $K = 2$, the second excited shell is the terminal zero element, so $r_2 = 1/[p(p-1)]$ and $c_{\mathrm{G}} = (p+1)/(p-1)$. For $K > 2$, the second excited shell

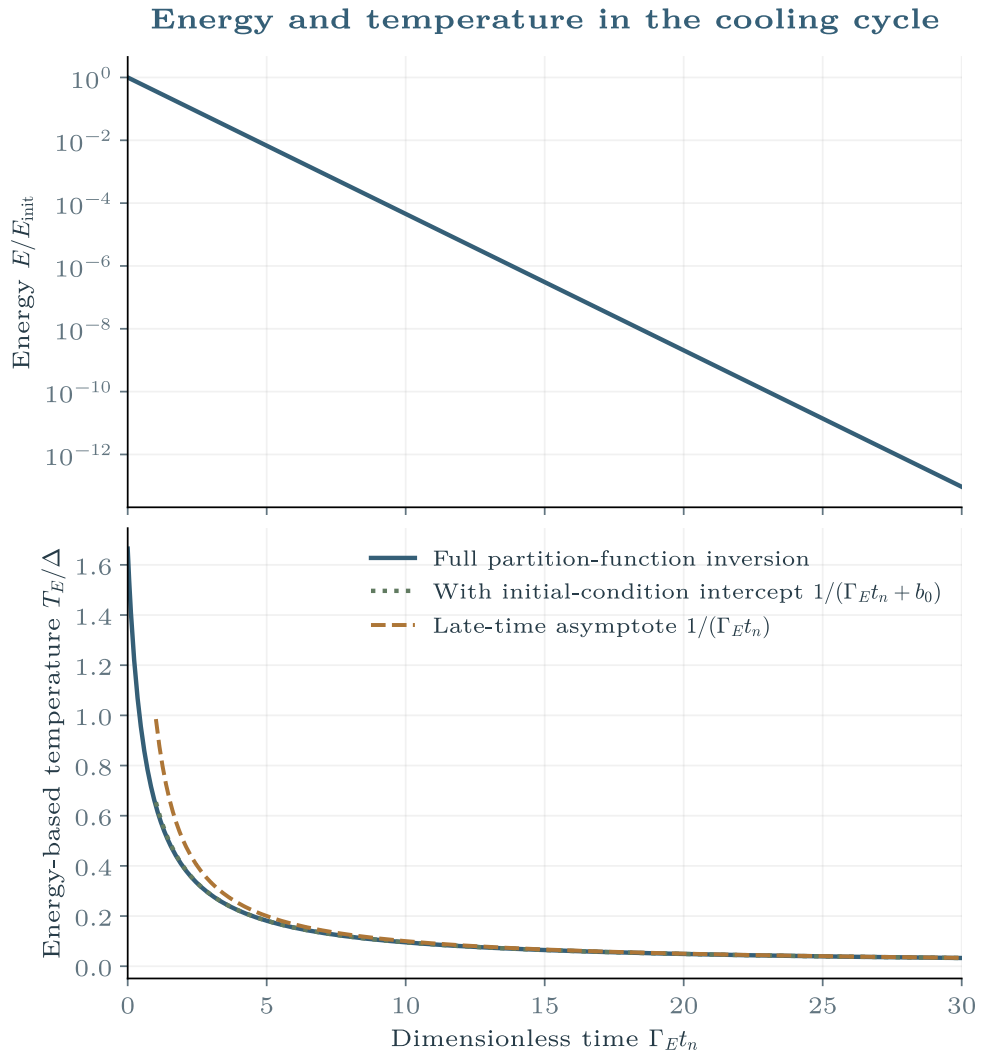


Figure 4-2. **Energy decrease and temperature reading.** Using the same channels as in Figure 4-1, take the environment and restoration preparations to zero temperature. Time is counted from the state after the first cooling cycle. All positive-energy modes have the same exit share, and energy decays exactly as $E_n = E_{\mathrm{init}} e^{-\Gamma_E t_n}$. Temperature is obtained by inverting the full finite partition function. The horizontal axis is the dimensionless time $\Gamma_E t_n$, and at late times $T_E/\Delta \sim 1/(\Gamma_E t_n)$. The solid curve uses the full spectrum; the dashed curve is the leading inverse-time law. The dotted curve includes the initial-condition intercept $b_0 = \log(\Delta d_1/(d_0 E_{\mathrm{init}}))$, corresponding to the late-time correction in the next section. The system starts in the Gibbs state at $\beta \log p = 0.2$ and then completes its first cooling cycle.

is an ordinary shell, giving $r_2 = 1/p^2$ and $c_{\mathrm{G}} = 1$. The truncation boundary thus enters the first correction to temperature calibration.

Meanwhile, the positive-energy modes of the preceding section have rates $\Gamma_\nu = -\log(1-\epsilon_\nu)/\tau_{\mathrm{tot}}$. Combining slow modes with the same rate into A_E, write the exact energy curve as

$$\begin{aligned} E_n &= A_E e^{-\Gamma_E t_n}(1+\eta_n), \\ \eta_n &= \sum_{\substack{E_\nu>0 \\ \Gamma_\nu>\Gamma_E}} \frac{E_\nu x_\nu(0)}{A_E} e^{-(\Gamma_\nu-\Gamma_E)t_n} \geq 0. \end{aligned} \tag{4.27}$$

If no faster mode is occupied, $\eta_n = 0$. Substituting this curve into the temperature expansion and setting $b_0 = \log(\Delta r_1/A_E)$ gives

$$\boxed{\begin{aligned} \Delta\beta_E(n) = \Gamma_E t_n + b_0 - \log(1+\eta_n) \\ + c_{\mathrm{G}}\frac{E_n}{\Delta} + O\left(\frac{E_n^2}{\Delta^2}\right). \end{aligned}} \tag{4.28}$$

The effect of subleading slow modes on the energy curve appears in the temperature reading as $-\log(1+\eta_n)$, while the finite partition function corrects the calibration through $c_{\mathrm{G}}E_n/\Delta$. The leading decay rate of the former is the smallest positive rate difference $\Gamma_\nu - \Gamma_E$ among occupied modes; the latter decays at rate Γ_E. Which contribution dominates depends on the actual spectrum, and coefficients must also be compared when the orders coincide. Retaining the full logarithm also includes higher-order terms of this spectral correction.

The constant b_0 retains the initial energy weight. Since $T_E = 1/\beta_E$, even when only one energy decay rate remains, finite-time deviations from the inverse-time law can be examined using the shifted hyperbola:

$$\frac{\Delta}{\Gamma_E t_n + b_0} = \frac{\Delta}{\Gamma_E t_n} - \frac{\Delta b_0}{(\Gamma_E t_n)^2} + O(t_n^{-3}). \tag{4.29}$$

The initial-condition intercept, actual energy spectrum, and temperature calibration separately affect the finite-time curve. Away from the low-temperature regime, the full partition function should be inverted directly.

As energy decreases, auxiliary outputs from each cycle retain the corresponding exchange information. Energy flow determines how the system cools; the outputs and their correlations further explain how the system continually loses its dependence on the initial state while the complete joint evolution remains reversible.

§5 Irreversibility of Thermalization

Thermalization changes the energy and internal composition of the system and carries distinctions between processes into auxiliary outputs. Once old outputs depart, the system continues exchanging with independently prepared fresh environments and gradually loses its dependence on the initial state. Retaining all outputs and their correlations in the joint description leaves the evolution reversible. The direction of the system's approach to thermal equilibrium follows from the entropy balance of the same complete cycle.

§5.1 Exchange Records

Denote the system and full auxiliary space after one cycle by S and $\mathcal{E}$. Their final joint state and system marginal are

$$\begin{aligned} \Omega_n^{\text{out}} &= \mathbb{U}_{\text{tot}}(\rho_n \otimes \eta_{\text{full}})\mathbb{U}_{\text{tot}}^\dagger, \\ \rho_{n+1} &= \text{Tr}_{\mathcal{E}}\, \Omega_n^{\text{out}}. \end{aligned} \tag{5.1}$$

Exchange, return, and background restoration are all included in this final state. Coherent retrieval first returns the same auxiliary to its initial preparation, transferring back to the system the directional distinctions it carried during the process. Background restoration subsequently preserves background distinctions in that auxiliary's output labels. Records must therefore be assessed from the states left by the thermal environment and auxiliary at the end of the complete cycle [1].

Let ω_α be the normalized conditional environment state for process component α, and let $r = f(\alpha)$ be the recorded category. Let Π_r^{rec} be the orthogonal projections of the actual pointer, including the no-response outcome. A complete record of the selected categories satisfies

$$\Pi_r^{\text{rec}}\Pi_s^{\text{rec}} = \delta_{rs}\Pi_r^{\text{rec}}, \qquad \sum_r \Pi_r^{\text{rec}} = I_{\mathcal{E}}. \tag{5.2}$$

$$\boxed{\Pi_r^{\text{rec}}\omega_\alpha = \delta_{r,f(\alpha)}\omega_\alpha.} \tag{5.3}$$

These distinctions must also have nonzero output shares and must not be mixed by subsequent actions during storage. Extend the record projections by the identity on the remaining spaces, writing $\widehat{\Pi}_r^{\text{rec}}$. The storage condition is

$$U_{\text{post}}^\dagger \widehat{\Pi}_r^{\text{rec}} U_{\text{post}} = \widehat{\Pi}_r^{\text{rec}}. \tag{5.4}$$

This holds when old outputs no longer participate in system exchange and storage operations preserve the pointer labels [17]. Records are formed by the actual joint action, and their existence does not depend on whether anyone reads them. If conditional states are only partially distinguishable, the accessible information resolves only part of the distinctions, and their actual overlaps must be retained.

Thermal exchange gives a concrete example. Retaining the channel and incoming energy-level labels and comparing the outgoing level identifies exit and reverse formation and reveals the environment energy change. Restoration-auxiliary labels distinguish background endpoints before and after restoration. Reading only the

outgoing energy level without retaining the incoming label generally does not determine the energy change of that exchange. Record resolution is thus determined by the distinctions actually preserved and cannot be replaced by an arbitrary choice of operators or representation.

The coherence decay of Chapter 3 is also determined by overlaps of these final states. If a pure joint component is written as $\sum_i A_i|i\rangle|\eta_i\rangle$, where $|\eta_i\rangle$ are conditional environment states, then

$$\langle i|\rho'|j\rangle = A_i\overline{A_j}\langle\eta_j|\eta_i\rangle. \tag{5.5}$$

Orthogonal outputs remove the corresponding cross terms from the system marginal; coincident outputs preserve coherence. Departed records and their correlations with the system and other outputs remain part of the joint state. Local information loss and preservation of joint information thus occur within the same process.

§5.2 Entropy and Thermal Equilibrium

First fix a finite positive inverse temperature β, take restoration shares $g = \pi$, and hold all other cycle data fixed. The system approaches the full Gibbs state $\rho_\beta = e^{-\beta H_K}/Z_K$ obtained in Chapter 2. We use von Neumann entropy and quantum relative entropy [18],

$$\begin{aligned} S(\rho) &= -\operatorname{Tr}(\rho\log\rho), \\ D(\rho\|\sigma) &= \operatorname{Tr}[\rho(\log\rho - \log\sigma)]. \end{aligned} \tag{5.6}$$

We use the convention $0\log 0 = 0$. At finite temperature, ρ_β has full rank, so every system state has finite relative entropy with respect to it. Substituting the logarithm of the Gibbs state yields

$$D(\rho\|\rho_\beta) = -S(\rho) + \beta\operatorname{Tr}(H_K\rho) + \log Z_K. \tag{5.7}$$

This includes both energy and the degree of internal mixing. The system's own entropy can decrease during cooling; the direction of approach to equilibrium is measured by deviation from the same thermal state.

The complete cycle is completely positive, trace-preserving, and preserves ρ_β. The data-processing inequality for relative entropy [19] therefore gives

$$\begin{aligned} D(\rho_{n+1}\|\rho_\beta) &= D(\mathcal{F}_\pi\rho_n\|\mathcal{F}_\pi\rho_\beta) \\ &\le D(\rho_n\|\rho_\beta). \end{aligned} \tag{5.8}$$

Let $D_n = D(\rho_n\|\rho_\beta)$. Expanding this quantity as above and using the complete energy balance of Chapter 4, the relaxation entropy production per cycle is

$$\boxed{\begin{aligned} \Sigma_n &= D_n - D_{n+1} \\ &= S(\rho_{n+1}) - S(\rho_n) \\ &\quad - \beta[Q_T(n) + Q_R(n)] \ge 0. \end{aligned}} \tag{5.9}$$

This relation includes energy flow in both thermal exchange and background restoration. Retaining only one contribution no longer gives the change of the full state relative to thermal equilibrium.

The common-restoration share of Chapter 2 also establishes strictness. Write $\sigma = \sigma_\pi^{\mathrm{sh}}$ and use $\mathcal{F}_\pi(A) = c\,\mathrm{Tr}(A)\sigma + (1-c)\mathcal{G}_\pi(A)$, where $c > 0$ and $\mathcal{G}_\pi$ is completely positive and trace-preserving. Joint convexity of relative entropy [19] first gives

$$\begin{aligned} D(\mathcal{F}_\pi\rho\|\rho_\beta) &= D\big(c\sigma + (1-c)\mathcal{G}_\pi\rho \\ &\qquad \|c\sigma + (1-c)\mathcal{G}_\pi\rho_\beta\big) \\ &\le (1-c)D(\mathcal{G}_\pi\rho\|\mathcal{G}_\pi\rho_\beta) \\ &\le (1-c)D(\rho\|\rho_\beta). \end{aligned} \tag{5.10}$$

The last step again uses data processing. Hence,

$$\boxed{\begin{aligned} D_n &\le (1-c)^n D_0, \\ \Sigma_n &\ge cD_n \ge 0. \end{aligned}} \tag{5.11}$$

Relative entropy decreases strictly whenever the system is not yet at thermal equilibrium. This statement applies to the full density operator, including occupations and coherences. The constant c gives a uniform estimate; actual decay rates remain determined by the modes in Chapter 3.

Summing over cycles cancels intermediate terms:

$$\sum_{n=0}^{N-1} \Sigma_n = D_0 - D_N \longrightarrow D_0. \tag{5.12}$$

For a fixed environment, a finite initial deviation contributes only a finite accumulated relaxation entropy production. Denoting nonequilibrium free energy at this temperature by $\mathscr{F}_\beta(\rho) = \mathrm{Tr}(H_K\rho) - \beta^{-1}S(\rho)$, the same result is

$$\begin{aligned} D(\rho\|\rho_\beta) &= \beta[\mathscr{F}_\beta(\rho) - \mathscr{F}_\beta(\rho_\beta)], \\ \mathscr{F}_\beta(\rho_n) - \mathscr{F}_\beta(\rho_{n+1}) &= \beta^{-1}\Sigma_n. \end{aligned} \tag{5.13}$$

When the environment temperature changes, the target thermal state moves as well. A nonnegative Σ_n can still be defined relative to the target ρ_{β_n} of each cycle, where $\beta_n = \beta_{\mathrm{B},n}$. Expanding the Gibbs-state logarithms for a moving target gives

$$\begin{aligned} &D(\rho_{n+1}\|\rho_{\beta_{n+1}}) - D(\rho_n\|\rho_{\beta_n}) \\ &\quad = -\Sigma_n + (\beta_{n+1} - \beta_n)E_{n+1} \\ &\qquad + \log Z_K(\beta_{n+1}) - \log Z_K(\beta_n). \end{aligned} \tag{5.14}$$

Thermalization reduces the deviation from the current equilibrium state in each cycle, while cooling changes the next target. Together they determine the actual lag. This relation applies at finite temperature. The zero-temperature reference state is not full rank, so the zero-temperature cooling curves of Chapter 4 continue to be calculated from the actual energy equations.

§5.3 Entropy Balance of the Environment

Once the system reaches thermal equilibrium, relaxation entropy production vanishes, yet the restoration auxiliary still changes from a pure state into an output carrying correlations. The system's relaxation balance must therefore be related to changes in environment preparations. Treating all auxiliaries of one cycle as a whole,

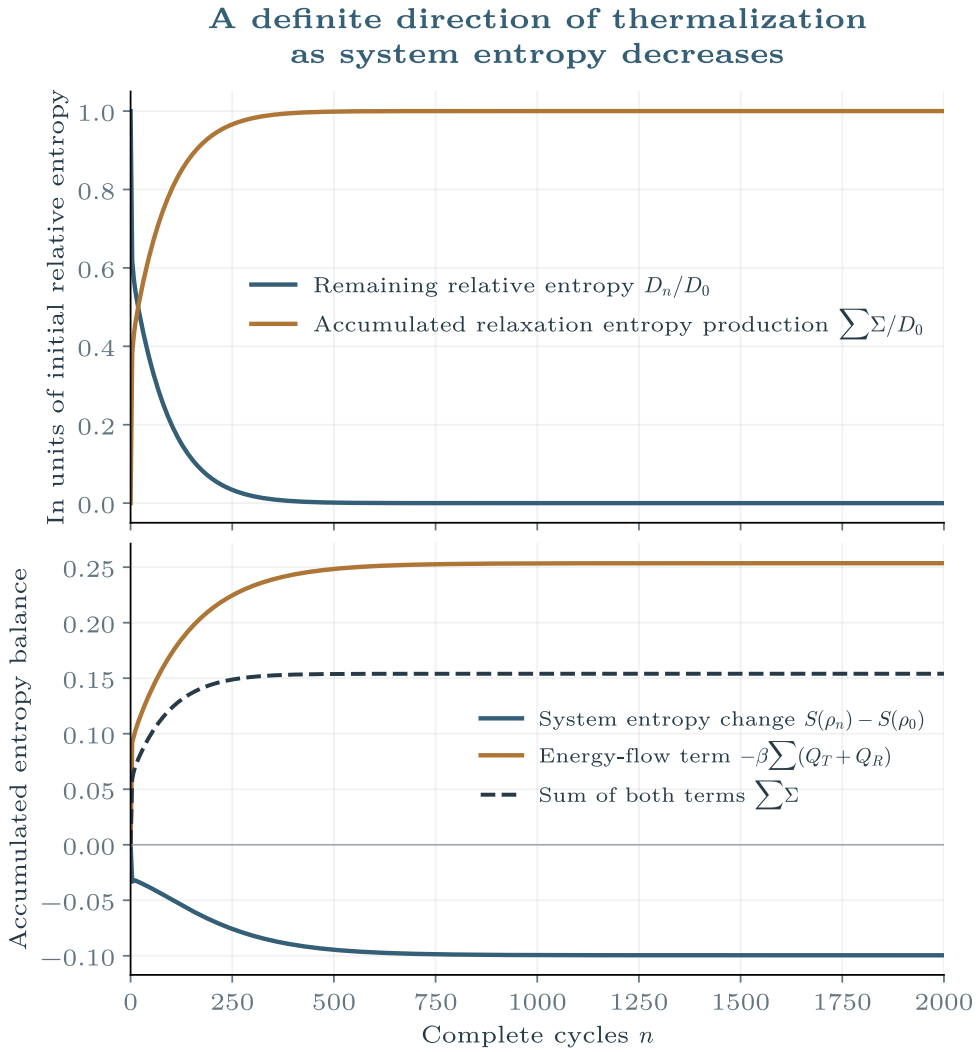


Figure 5-1. **Entropy change and approach to equilibrium.** For the 49-dimensional complete cycle of Chapter 4, fix the environment at $\beta \log p = 2.2$ and prepare the system in the Gibbs state at $\beta_0 \log p = 0.2$. The upper panel shows system relative entropy and accumulated relaxation entropy production, the latter equal to the decrease of relative entropy. The lower panel separates the system entropy change and complete energy-flow contribution. The system's entropy decreases during cooling, but $\Delta S_S - \beta \sum(Q_T + Q_R)$ is nonnegative. The curves propagate the actual occupations without imposing instantaneous thermal equilibrium.

the initial state is $\Omega_{\text{in}} = \rho \otimes \eta_{\mathcal{E}}$. Joint unitary evolution preserves its spectrum, so $S(\Omega_{\text{out}}) = S(\rho) + S(\eta_{\mathcal{E}})$. The definition of mutual information directly gives

$$\boxed{\begin{aligned} \Delta S_S + \Delta S_{\mathcal{E}} &= I(S:\mathcal{E})_{\text{out}}, \\ I(S:\mathcal{E})_{\text{out}} &= S(\rho') + S(\eta'_{\mathcal{E}}) - S(\Omega_{\text{out}}). \end{aligned}} \tag{5.15}$$

Joint entropy is unchanged, while the increase in the sum of the marginal entropies equals the correlations created. Information can reside in an individual output or in correlations between the system and outputs or among different outputs. Nonzero mutual information signifies correlation; whether it constitutes a storable record is still determined by the criteria of Section 5.1.

The two environment preparations in the complete cycle differ, and so do the sources of their entropy balances. The thermal environment has a Gibbs preparation at the same temperature within each direct-sum block, and thermal exchange preserves the block weights. The blockwise normalization constants in $\log \eta_T$ therefore cancel between the final and initial states. Since the environment energy change is $-Q_T$, we obtain [1,20,21]

$$\begin{aligned} D(\eta'_T \| \eta_T) &= -\Delta S_T - \beta Q_T, \\ \Delta S_{\text{th}}^{\text{sys}} - \beta Q_T &= I_T + D(\eta'_T \| \eta_T) \geq 0. \end{aligned} \tag{5.16}$$

Here I_T is the mutual information between the system and thermal environment at the end of thermal exchange. The two terms on the right account for correlations created during exchange and changes in the thermal environment relative to its initial preparation.

The actual input of the restoration auxiliary is the pure state P_g. Denote the background block after thermal exchange by $y\xi_{\mathcal{C}}$, where y is the total background share and $\xi_{\mathcal{C}}$ is normalized. In the interaction picture of restoration, $|e_b, g\rangle \mapsto \sum_a \sqrt{\pi_a} |e_a, r_{ab}\rangle$ transfers the input matrix structure into distinct auxiliary blocks. Its final state and entropy are therefore

$$\begin{aligned} \eta'_R &= (1-y) P_g \ \oplus \bigoplus_a y \pi_a \xi_{\mathcal{C}}, \\ S(\eta'_R) &= h_2(y) + y[\mathsf{H}(\pi) + S(\xi_{\mathcal{C}})]. \end{aligned} \tag{5.17}$$

Here $h_2(y) = -y \log y - (1-y)\log(1-y)$ and $\mathsf{H}(\pi) = -\sum_a \pi_a \log \pi_a$. For $y = 0$, zero-weight terms are understood by continuity. The complete free evolution further conjugates the final auxiliary state by $e^{-i\tau_{\text{tot}} H_R}$, leaving its spectrum, entropy, and energy unchanged. Restoration establishes the prescribed background distribution in the system while retaining the original background distinctions in the auxiliary.

To quantify the preparation change, take the full-rank comparison state $\omega_R = e^{-\beta H_R}/Z_R$ at the same finite temperature and define

$$C_R = D(P_g \| \omega_R) - D(\eta'_R \| \omega_R). \tag{5.18}$$

The actual input remains P_g. Expanding the two relative entropies and using the restoration-auxiliary energy change $-Q_R$ gives

$$C_R = S(\eta'_R) + \beta Q_R. \tag{5.19}$$

The independent pure auxiliary input in the restoration stage also gives $\Delta S_R^{\mathrm{sys}} = I_R - S(\eta'_R)$. Adding thermal exchange and restoration, with the system's unitary phase from the coherent round trip leaving its entropy unchanged, yields the complete balance

$$\boxed{\Sigma_n + C_R = I_T + D(\eta'_T \| \eta_T) + I_R \geq 0.} \tag{5.20}$$

The quantities I_T and I_R belong to the two separate stages. Their sum also includes final correlations between the thermal environment and restoration auxiliary, so it cannot simply be replaced by the final mutual information between the system and the entire environment. The term C_R describes the change in auxiliary preparation relative to the same comparison state; its sign is determined by the actual states and energy flows. Repreparing auxiliaries involves information reset and its physical cost [22], while the actual work required depends on the preparation operation.

When the system is already in ρ_β, thermal exchange preserves the joint input, giving $I_T = D(\eta'_T \| \eta_T) = Q_T = 0$. Restoration preserves the system thermal state, with $Q_R = \Sigma_n = 0$. Nevertheless, the restoration auxiliary changes from a pure state into an output carrying correlations. Setting $y_* = Z_C/Z_K$ and $\xi_C = \sigma_\pi^{\mathrm{sh}}$, the preceding relation gives

$$C_R = I_R = h_2(y_*) + 2y_* \mathsf{H}(\pi) > 0. \tag{5.21}$$

These balances refer to the joint realization without an additional label-copying device. If labels are actually copied or stored, the relevant memory units also belong to the environment and must enter the same joint entropy relation.

§5.4 Forward and Reverse Processes and the Direction of Time

The direction of mean entropy production can also be expressed through recordable exchange trajectories. Forward-to-reverse probability ratios are central quantities in heat-exchange fluctuation relations [23]. Take an initial state diagonal in the common mode basis $\{\nu, e_a\}$, with strictly positive initial shares in every mode, and retain the initial mode, incoming and outgoing thermal-exchange labels, and restoration labels. The conditional operators are diagonal operators or individual matrix units, so each recorded history tracks mode continuation and transfer. Summing all labels recovers the established diagonal occupation dynamics. Below, records are grouped by mode path, with unresolved auxiliary labels within the same path already summed.

Let B_{ji} be the thermal-exchange share from mode i to j, and R_{kj} the subsequent restoration share from j to k. Their nonzero entries are

$$\begin{aligned} \mathsf{B}_{e_a,\nu} &= \epsilon_{a\nu}, & \mathsf{B}_{\nu,e_a} &= f_{\nu a}, \\ \mathsf{B}_{\nu,\nu} &= 1 - \epsilon_\nu, & \mathsf{B}_{e_a,e_a} &= 1 - \sum_\nu f_{\nu a}, \\ \mathsf{R}_{e_b,e_a} &= \pi_b, & \mathsf{R}_{\nu,\nu} &= 1. \end{aligned} \tag{5.22}$$

Every column sums to one, and other entries vanish. Let $q_i = e^{-\beta E_i}/Z_K$ be the full Gibbs shares. The forward-to-reverse exchange ratio and same-temperature restoration shares give, respectively,

$$\mathsf{B}_{ji} q_i = \mathsf{B}_{ij} q_j, \qquad \mathsf{R}_{kj} q_j = \mathsf{R}_{jk} q_k. \tag{5.23}$$

A forward cycle performs exchange before restoration, with occupation kernel $\mathsf{P}_{\mathrm{F}} = \mathsf{RB}$. To compare trajectories in the opposite order, take the process $\mathsf{P}_{\mathrm{R}} = \mathsf{BR}$, which restores before exchanging. Each stage separately satisfies detailed balance; their product need not satisfy detailed balance in the same stage order.

For a one-cycle path $i \to j \to k$ with nonzero share, the paired reverse-order path is $k \to j \to i$. Multiplying the two stage ratios cancels the intermediate mode weight:

$$\frac{\mathsf{B}_{ji}\mathsf{R}_{kj}}{\mathsf{R}_{jk}\mathsf{B}_{ij}} = \frac{q_k}{q_i} = e^{\beta(E_i - E_k)}. \tag{5.24}$$

The quantity $E_i - E_k$ is the combined energy released to the auxiliaries through thermal exchange and restoration.

Let ω contain the mode sequence and intermediate endpoints over N cycles. The forward process starts from p_0 and ends in p_N. The reverse-order comparison starts from p_N, traverses the stages in the opposite order, and follows the paired path $\bar{\omega}$. Endpoint shares and conditional transition shares give

$$\begin{aligned} \mathbb{P}_{\mathrm{F}}[\omega] &= p_0(i_0) \prod_{n=0}^{N-1} \mathsf{B}_{j_n i_n} \mathsf{R}_{i_{n+1} j_n}, \\ \mathbb{P}_{\mathrm{R}}[\bar{\omega}] &= p_N(i_N) \prod_{n=0}^{N-1} \mathsf{R}_{j_n i_{n+1}} \mathsf{B}_{i_n j_n}. \end{aligned} \tag{5.25}$$

The logarithmic path-probability ratio is then

$$\boxed{\begin{aligned} \sigma[\omega] &= \log \frac{\mathbb{P}_{\mathrm{F}}[\omega]}{\mathbb{P}_{\mathrm{R}}[\bar{\omega}]} \\ &= \log \frac{p_0(i_0)}{p_N(i_N)} + \beta(E_{i_0} - E_{i_N}). \end{aligned}} \tag{5.26}$$

The forward and reverse shares have matching support. Grouping paired paths by σ yields the detailed and integral fluctuation relations [24,25]

$$\begin{aligned} \frac{\mathcal{P}_{\mathrm{F}}(s)}{\mathcal{P}_{\mathrm{R}}(-s)} &= e^{s}, \\ \left\langle e^{-\sigma} \right\rangle_{\mathrm{F}} &= 1. \end{aligned} \tag{5.27}$$

The reverse-process variable is defined with the opposite sign on paired paths, and its endpoint comparison still uses p_N, p_0 above. The denominator is the distribution for the specified reverse-order process; it cannot in general be replaced by the distribution of negative values in the same forward process. Under the forward path average, the initial and final modes are distributed according to p_0, p_N, so

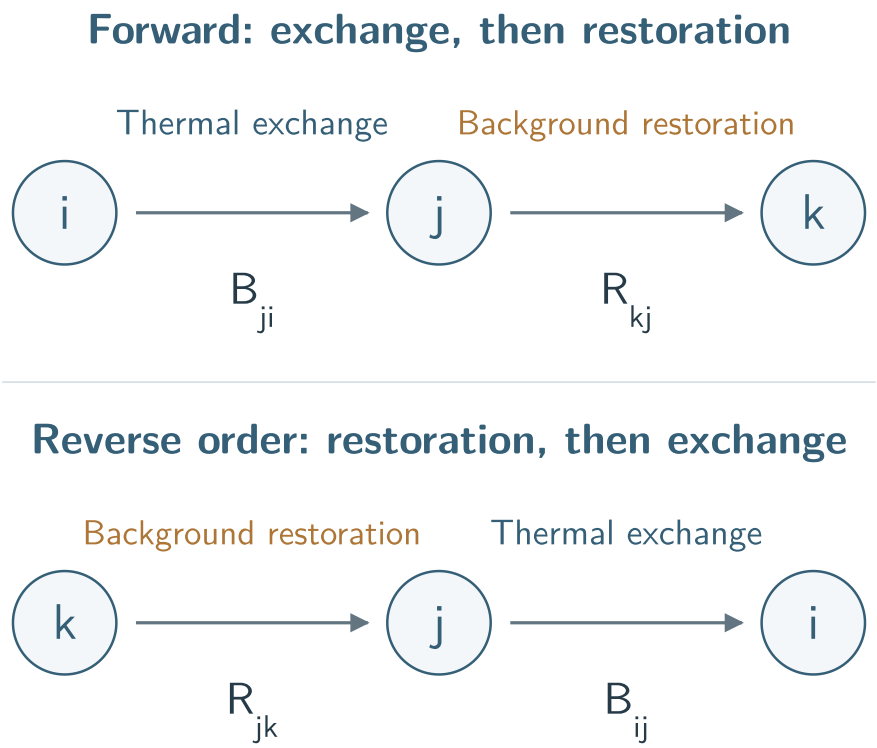


Figure 5-2. **Forward and reverse comparison of exchange trajectories.** The upper path performs thermal exchange followed by background restoration; the lower path pairs it with the opposite order of the two stages. Multiplication of the stagewise Gibbs-weight ratios cancels the intermediate weight. This comparison keeps the same temperature and stage transition shares. It differs from returning all the original auxiliary outputs with their correlations and applying the inverse joint unitary.

$$\begin{aligned}\langle\sigma\rangle_{\mathrm{F}} &= S(p_N) - S(p_0) + \beta[E_S(0) - E_S(N)] \\ &= D(p_0\|q) - D(p_N\|q) \\ &= \sum_{n=0}^{N-1} \Sigma_n \geq 0.\end{aligned} \tag{5.28}$$

A single trajectory can have negative σ, while the mean direction remains consistent with relaxation of the full state. This relation concerns system relaxation along diagonal trajectories; changes in auxiliary preparations are still accounted for by Section 5.3. For general coherent initial states, the quantum-relative-entropy result of Section 5.2 applies directly.

Under the environment refreshing conditions of Section 1.2, all cycles can still be included in one joint unitary evolution. Retain every output and its coherent correlations over N cycles, including the actual recording units. Denote the initial joint state by $\Omega^{(0)}$ and the joint evolution by $\mathbb{V}_N = \mathbb{U}_N \cdots \mathbb{U}_1$. Then

$$\mathbb{V}_N^\dagger \Omega^{(N)} \mathbb{V}_N = \Omega^{(0)}. \tag{5.29}$$

Realizing this recovery requires the actual final joint state and inverse actions in the opposite order. Continuing to supply independent fresh auxiliaries does not implement that inverse process. Chapter 2 shows that the trace distance between any two distinct system initial states contracts by a factor no greater than $1 - c$. Trace-norm contraction of subsequent system channels [26] prevents universal recovery of that distance, so no physical system-only inverse channel works for every initial state. The local time direction of thermalization is jointly determined by the complete dynamics and environment refreshing conditions. Preserved records allow earlier distinctions to remain readable.

§6 Tests of the Thermalization Mechanism

Testing the thermalization mechanism requires connecting a single energy exchange to the relaxation that follows. Once the exchange shares, environment preparations, and interaction durations are fixed, the same complete cycle jointly constrains the equilibrium distribution, occupation relaxation, coherence decay, and energy output. Reading these quantities separately tests whether they can be explained by one set of dynamical parameters. Changing whether old environments return further tests the environmental conditions on which repeated thermalization depends.

§6.1 Exchange Statistics

Separately prepare matter mode ν and background endpoint e_a, and repeat one complete cycle with the same environment preparation and exchange action. Following the record criteria of Chapter 5, retain incoming and outgoing thermal-environment labels and count exit and reverse-formation events in the thermal-exchange stage. Let N_ν, N_{e_a} be the total trial counts for the two initial preparations. The corresponding event counts give

$$\widehat{\epsilon}_{a\nu} = \frac{N_{\nu\to e_a}}{N_\nu}, \qquad \widehat{f}_{\nu a} = \frac{N_{e_a\to\nu}}{N_{e_a}}. \tag{6.1}$$

The denominators include all trials, including no-response outcomes and transitions into other channels. These are thermal-exchange events and cannot be identified by substituting system background labels measured after restoration. Normalizing only within the selected channel's sample would omit its preparation weight from the measured share.

For independent repeated preparations and sufficiently large samples, the counting shares approach the actual exchange shares of Chapter 2. Independently calibrated channel weights, incoming energy-level distributions, and exchange angles predict

$$\begin{aligned} \epsilon_{a\nu} &= a_{a\nu} t_u^{(a\nu)} \sin^2 \alpha_{a\nu}, \\ f_{\nu a} &= a_{a\nu} t_v^{(a\nu)} \sin^2 \alpha_{a\nu}. \end{aligned} \tag{6.2}$$

Their ratio cancels the common interaction strength, leaving the relation between energy difference and environment temperature:

$$\boxed{\log \frac{\epsilon_{a\nu}}{f_{\nu a}} = -\beta(E_a - E_\nu).} \tag{6.3}$$

Repeating the measurement for different energy differences should give the same β under a common-temperature preparation. Zero-energy-difference channels test equality of the two shares but cannot alone determine temperature. Raw forward and reverse event counts during equilibrium also include the respective initial occupations. Their equality indicates zero net flow; their ratio cannot be identified directly with the conditional-share ratio above.

Restoration-auxiliary labels can independently determine the background distribution. Conditional on trials that actually enter restoration, count the output endpoints to obtain $\widehat{g}_a$. The restoration distribution required at the common temperature is

$$\pi_a = \frac{e^{-\beta E_a}}{Z_{\mathcal{C}}}, \qquad Z_{\mathcal{C}} = \sum_b e^{-\beta E_b}. \tag{6.4}$$

Comparing $\widehat{g}$ with π at the temperature determined by the exchange ratios directly tests the same-temperature restoration condition of Section 2.3.

§6.2 Joint Tests of Occupations and Coherence

Substituting the measured shares into the occupation matrix of Chapter 3, with time counted from the restored endpoint used there, predicts occupations over many cycles. The required parameters are

$$\begin{aligned} \epsilon_\nu &= \sum_a \epsilon_{a\nu}, \quad & b_\nu &= \sum_a g_a f_{\nu a}, \\ f_\Sigma &= \sum_\nu b_\nu, \quad & \frac{x_\nu^*}{y_*} &= \frac{b_\nu}{\epsilon_\nu}. \end{aligned} \tag{6.5}$$

The same matrix gives equilibrium shares and the decay factors of all occupation deviations. When $N \geq 2$ and all matter modes have the same exit share $\epsilon_\nu = \epsilon$, the total-share deviation and internal-composition deviations satisfy

$$
\begin{aligned}
y_n - y_* &= \lambda_{\mathrm{tot}}^n (y_0 - y_*), \\
z_{\nu,n} &= \lambda_{\mathrm{comp}}^n z_{\nu,0}, \\
\lambda_{\mathrm{tot}} &= 1 - \epsilon - f_\Sigma, \qquad \lambda_{\mathrm{comp}} = 1 - \epsilon.
\end{aligned}
\tag{6.6}
$$

Here $z_\nu = x_\nu - (b_\nu / f_\Sigma)(1 - y)$. Different initial preparations can excite the two kinds of deviations separately. Changes in composition among equal-energy modes do not appear in mean energy, so the corresponding mode occupations must still be read.

Coherence supplies another independent reading. Under the same conditions, prepare a superposition of ν, μ, evolve for n cycles, and measure at the final endpoint in the states

$$
|\pm_\theta\rangle = \frac{|\nu\rangle \pm e^{-i\theta}|\mu\rangle}{\sqrt{2}}.
\tag{6.7}
$$

Treat results outside this subspace as a third outcome and normalize by the total number of trials. The probability difference between the two projections is

$$
\begin{aligned}
C_\theta(n) &= P_+(n) - P_-(n) \\
&= 2\,\mathrm{Re}[e^{-i\theta}\rho_{\nu\mu}(n)], \\
C_0(n) + iC_{\pi/2}(n) &= 2\rho_{\nu\mu}(n).
\end{aligned}
\tag{6.8}
$$

Measurements at the two phase settings therefore determine the real and imaginary parts of this matrix element. Retaining only samples that remain in the two-mode subspace would additionally divide the reading by $x_\nu + x_\mu$, changing the decay law under test.

For the complete cycle of Chapter 3 and initial state $|+_0\rangle$,

$$
\boxed{
\begin{aligned}
C_\theta(n) &= \gamma_{\nu\mu}^n \cos[n(\varphi_\nu - \varphi_\mu) - \theta], \\
\gamma_{\nu\mu} &= 1 - \sum_a \frac{\epsilon_{a\nu}}{1 + \cos\alpha_{a\nu}} \\
&\quad - \sum_a \frac{\epsilon_{a\mu}}{1 + \cos\alpha_{a\mu}}.
\end{aligned}
}
\tag{6.9}
$$

The phase difference comes jointly from free evolution and coherent retrieval over the complete cycle, while the coherence envelope and exit shares are constrained by the same exchange angles. Knowing only a total exit share is insufficient. Once channel-resolved shares and exchange angles are independently determined, the coherence curve provides an additional test.

Figure 6-1 uses the seven-dimensional example of Chapter 3: five zero-energy matter modes exchange only with the zero-energy background endpoint, each with weight $2/15$. The preparation weight for no thermal exchange is $1/3$, and restoration shares are $(4/5, 1/5)$. Varying the common angle $0 < \alpha < \pi/2$ gives $\epsilon = \sin^2\alpha/15$ and $f_\Sigma = 4\epsilon$. Eliminating α requires the three per-cycle factors to lie on the same set of curves:

$$
\begin{aligned}
\lambda_{\mathrm{tot}} &= 1 - 5\epsilon, \\
\lambda_{\mathrm{comp}} &= 1 - \epsilon, \\
\gamma_{\nu\mu} &= 1 - \frac{2}{15}\big(1 - \sqrt{1 - 15\epsilon}\big).
\end{aligned}
\tag{6.10}
$$

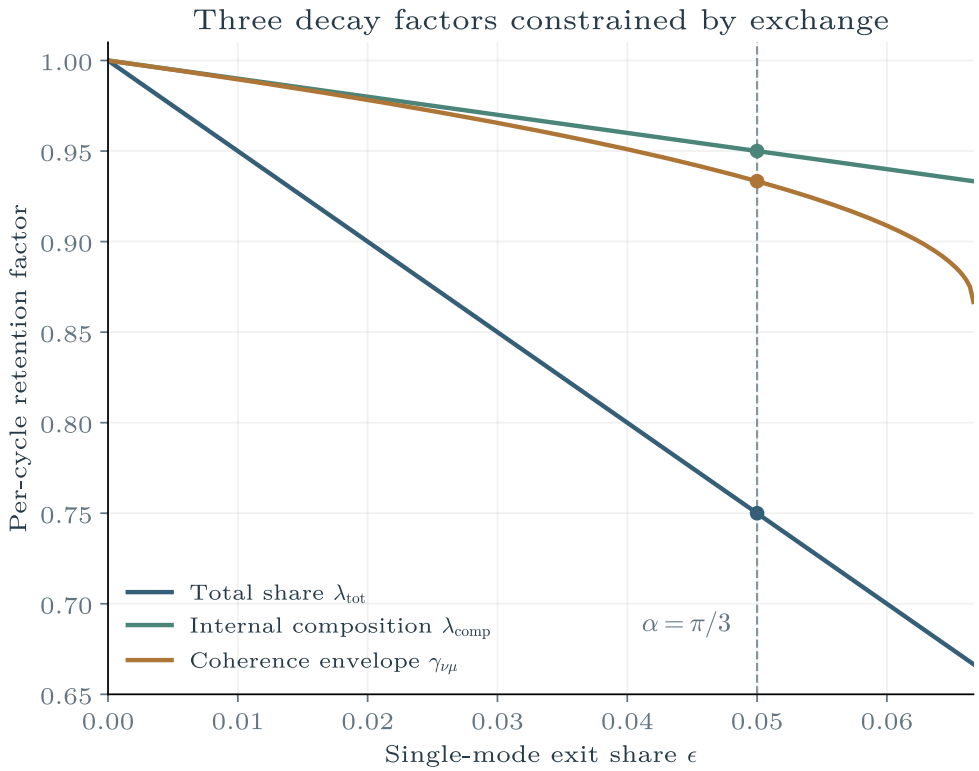


Figure 6-1. **Three decay factors constrained by the same exchange share.** The horizontal axis is the independently measured single-mode exit share; the vertical axis is the per-cycle retention factor. All three curves are determined by the same complete cycle. Marked points correspond to $\alpha = \pi/3$: the total-share factor is 3/4, the composition factor 19/20, and the coherence factor 14/15. These are theoretical relations for the specified cycle, not experimental data.

Conversion to physical time uses $\Gamma = -\log\lambda/\tau_{\mathrm{tot}}$, taking $\lambda = \gamma_{\nu\mu}$ for the coherence envelope. The total elapsed duration must be measured independently. If changing the exchange angle also changes the cycle period, a change in the per-cycle factor cannot be identified directly with a change in the decay rate per unit time.

§6.3 Cooling and Entropy Balance

Energy tests connect system changes to actual environment outputs. Incoming and outgoing thermal-environment energies and the initial and final restoration-auxiliary energies separately determine the energy transferred to the system in the two stages. If the mean system energy is E_n, one can independently compare

$$\begin{aligned} Q_T(n) &= -\Delta\langle H_T\rangle_n, \\ Q_R(n) &= -\Delta\langle H_R\rangle_n, \\ E_{n+1} - E_n &= Q_T(n) + Q_R(n). \end{aligned} \tag{6.11}$$

In this cycle, coherent round trips and free evolution contribute no net system energy change [1]. Any additional driving or leakage in a physical realization requires the corresponding energy flows to be included. Thermal-exchange and restoration outputs can be read stage by stage in independent repetitions with the same prepa-

ration, avoiding changes to the original cycle from additional intermediate measurements.

The cooling curve of Chapter 4 can be tested in the same way. Use its zero-temperature environment and restoration to the zero-energy background, so positive-energy modes can only exit and have the same exit share $\epsilon > 0$. Starting from the state after one completed cooling cycle, with initial mean energy $E_{\mathrm{init}} > 0$,

$$\begin{aligned} E_n &= E_{\mathrm{init}}(1-\epsilon)^n, \\ \Gamma_E &= -\frac{\log(1-\epsilon)}{\tau_{\mathrm{tot}}}, \\ T_E(t) &\sim \frac{\Delta}{\Gamma_E t}, \qquad \Delta = \log p. \end{aligned} \tag{6.12}$$

After determining ϵ from single-cycle exchange, compare the predicted energy curve with system and environment energy readings. The quantity T_E is obtained from this energy through the full partition function.

Entropy adds a constraint beyond energy. Return to a fixed finite positive temperature with same-temperature restoration, reconstruct the system density operator from repeated trials with the same preparation, and compare it with the two independently measured energy flows:

$$\boxed{\begin{aligned} \Sigma_n &= S(\rho_{n+1}) - S(\rho_n) \\ &\quad - \beta[Q_T(n) + Q_R(n)] \\ &= D_n - D_{n+1} \geq 0. \end{aligned}} \tag{6.13}$$

For preparations that remain diagonal, all mode probabilities determine the system entropy. When coherence is present, off-diagonal structure must be included. Entropy calculated only from total energy-shell shares is generally insufficient to test this relation.

If actual restoration outputs are also read, the preparation change and complete balance of Chapter 5 give

$$\begin{aligned} C_R &= S(\eta'_R) + \beta Q_R, \\ \Sigma_n + C_R &= I_T + D(\eta'_T \| \eta_T) + I_R. \end{aligned} \tag{6.14}$$

These quantities must be determined from the joint system–auxiliary states at their respective stages.

§6.4 Environment Refreshing and Return

The preceding multi-cycle predictions use independent fresh environments. To test this condition, keep the single-cycle joint action fixed but retain and reintroduce the same thermal environment and restoration auxiliary, preserving their correlations between cycles. The next input is then generally no longer a product of the system and a prescribed fresh environment, so the full joint state must be propagated [27]. Effects of environmental correlations on successive collisions have been studied in specific collision models and their optical simulations [28,29].

A finite environment can also realize the fresh-auxiliary arrangement for a finite observation time. Prepare N mutually independent sets of complete auxiliaries, also independent of the initial system, each in the prescribed η_{full}. Use one set per cycle,

with old outputs no longer participating and unused auxiliaries left undisturbed. The first N cycles then give exactly $\rho_n = \mathcal{F}^n(\rho_0)$. Beyond this supply window, preparation and return conditions must be reconsidered. The comparison below instead retains the same auxiliaries from the first cycle onward.

This comparison requires the complete unitary action to be fixed, including its action on returning restoration outputs. We use the restoration rotation of the established cycle [1]:

$$\begin{aligned} U_{\mathrm{res}} &= \exp\Big[\frac{\pi}{2}(L - L^\dagger)\Big], \\ L &= \sum_b |Y_b\rangle\langle e_b, g|, \\ |Y_b\rangle &= \sum_a \sqrt{g_a}|e_a, r_{ab}\rangle. \end{aligned} \tag{6.15}$$

Here $|g\rangle$ is the initial restoration-auxiliary label and $|r_{ab}\rangle$ are its orthogonal output labels. For the zero-temperature preparation $g_a = \delta_{a0}$, we have $|Y_b\rangle = |e_0, r_{0b}\rangle$. This rotation specifies forward restoration, output return, and the action on the orthogonal complement.

Prepare the system initially in a positive-energy matter mode $|\nu\rangle$ that exchanges only with e_0, with channel weight a and exchange angle α. At zero temperature, the thermal input for this channel is $|u\rangle$. The interaction-picture retrieval phase is $w_\nu = e^{i(\varphi_\nu + \tau_{\mathrm{tot}} E_\nu)}$; together with the free phase it gives the net cycle phase of Chapter 3. Choose $w_\nu = 1$ for this example; other directions need not have the same phase. In the order system, thermal environment, restoration auxiliary, define

$$\begin{aligned} |A\rangle &= |\nu, u, g\rangle, \qquad |B\rangle = |e_0, v, g\rangle, \\ |C\rangle &= |e_0, v, r_{00}\rangle, \quad |D\rangle = |\nu, u, r_{00}\rangle. \end{aligned} \tag{6.16}$$

These four states span an invariant sector of the complete joint action. The coherent round trip within a stage still passes through intermediate states and returns to this sector after full retrieval. Thermal exchange rotates the pairs (A, B) and (D, C), while restoration gives $B \mapsto C$ and $C \mapsto -B$. All four states have the same total energy. Removing the common free phase and writing $c_\alpha = \cos\alpha$, $s_\alpha = \sin\alpha$, the single-cycle action is

$$\boxed{U_4 = \begin{pmatrix} c_\alpha & -s_\alpha & 0 & 0 \\ 0 & 0 & -c_\alpha & -s_\alpha \\ s_\alpha & c_\alpha & 0 & 0 \\ 0 & 0 & -s_\alpha & c_\alpha \end{pmatrix}}. \tag{6.17}$$

When all auxiliaries are refreshed each cycle, the system encounters this channel with share a each time. Once the positive-energy mode exits, it cannot be reformed by a zero-temperature environment. Its occupation is therefore

$$P_\nu^{\mathrm{new}}(n) = (1 - a\sin^2\alpha)^n. \tag{6.18}$$

When the same auxiliaries are retained, the initial share $1-a$ in other orthogonal thermal blocks never exchanges with this mode. The share a in the selected block evolves repeatedly within the four-state sector. Setting $v_n = U_4^n|A\rangle$ gives

$$\boxed{\begin{aligned} q_\nu(n) &= |\langle A|v_n\rangle|^2 + |\langle D|v_n\rangle|^2, \\ P_\nu^{\mathrm{ret}}(n) &= 1 - a + a q_\nu(n). \end{aligned}} \tag{6.19}$$

The first cycle is identical in both processes. In the second cycle, the four-state action gives conditional occupation $\cos^4\alpha + \sin^4\alpha$ of ν, and hence

$$\boxed{P_\nu^{\mathrm{ret}}(2) - P_\nu^{\mathrm{new}}(2) = a(2-a)\sin^4\alpha > 0.} \tag{6.20}$$

The difference is produced by reusing the same outputs. For $a = 0.4$ and $\alpha = 0.8$, the first three occupations with refreshed auxiliaries are approximately $0.7942, 0.6307, 0.5009$; with retained auxiliaries, they are $0.7942, 0.8002, 0.9998$, already showing pronounced return.

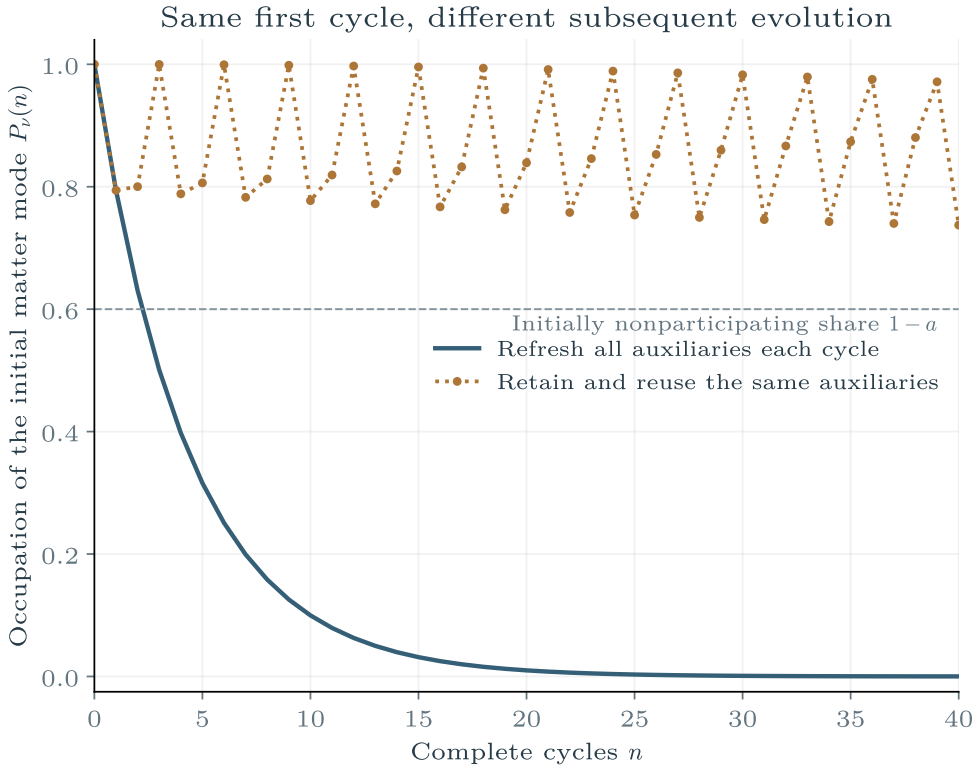


Figure 6-2. **The same action under different environmental conditions.** The system starts in positive-energy mode ν, with selected-channel parameters $a = 0.4$, $\alpha = 0.8$, and $w_\nu = 1$, and restoration to the zero-energy endpoint. The solid curve uses independent fresh auxiliaries each cycle. The dotted curve retains and repeatedly reuses the same thermal environment and restoration auxiliary; it is calculated directly from U_4^n, without measurements or removal of correlations between cycles. This configuration embeds in the full 49-dimensional cycle: each of the other 45 matter modes has channel weight 0.01, and the preparation weight for no thermal exchange is 0.15. The figure gives the exact result for this specified process.

More general return phases, finite temperatures, and auxiliary structures change the return curve and require the corresponding complete unitary action to be calculated. Absence of visible return during a finite observation time does not establish that the environment never returns.

Table 6-1 summarizes the relations between independent parameters and multi-cycle readings.

Table 6-1. **Parameters and test quantities of the same cycle.** Each row retains the temperature, initial-state, and channel conditions of its corresponding section.

Independently determined quantities	Results to compare	Relation tested
Channel weights, exchange angles, incoming energy-level distribution	Exit and formation shares and their ratio	Single-exchange strength and temperature
Exchange shares, restoration distribution	Full equilibrium occupations	Whether exchange and restoration have the same temperature
Single-cycle shares, restoration distribution, exchange angles, initial state, total elapsed duration	Occupation modes, coherence envelope	Whether one cycle explains different relaxation rates
Initial occupations, exit shares, auxiliary energy changes in both stages	System energy changes and cooling curve	Cooling dynamics and complete energy balance
Stage-resolved joint system–auxiliary states, energy flows	Relaxation entropy production and preparation change	Consistency of system balance and environment correlations
Single-cycle joint action, auxiliary retention arrangement	Multi-cycle differences between refreshing and return	Independent preparation and returning correlations

§7 Conclusion

Thermal equilibrium is established through sustained exchange between matter and the environment. Matter exit, renewed formation, and background restoration jointly change the overall occupations, while coherence evolves with distinctions in the full joint outputs. The complete cycle determines the equilibrium distribution, relaxation modes, and information balance of the system and environment together.

In a finite state space, fix the exchange action and a finite positive temperature, take the thermal environment and restoration distribution at the same temperature, and supply fresh auxiliaries independent of the joint state of the system and all old outputs, with old outputs not returning. If the preparation weight for no thermal exchange is $a_0 > 0$ and every matter mode has exit share $\epsilon_\nu > 0$, every initial state converges to the unique full thermal state:

$$\boxed{\begin{aligned} &\mathcal{F}_{\pi}^{n}(\rho_0) \longrightarrow \rho_\beta, \\ &\rho_\beta = \frac{e^{-\beta H_K}}{Z_K}, \\ &Z_K = \sum_{j=0}^{K} |S_j| e^{-\beta E_j}. \end{aligned}} \tag{7.1}$$

This result includes relaxation of the energy distribution, matter composition within shells, and coherence. Shell degeneracies enter the full partition function, and the restoration preparation enters the equilibrium condition. The temperature specified by forward-to-reverse exchange ratios must be preserved throughout the complete cycle. Equilibrium is a statistical balance of shares, while individual exchanges can continue.

For the propagation used here, which does not mix matter modes, the nonequilibrium modes of the same cycle determine the relaxation rates. After the first background restoration, a nonzero decay factor λ_r corresponds to the physical-time rate

$$\boxed{\Gamma_r = -\frac{\log|\lambda_r|}{\tau_{\text{tot}}}, \qquad 0 < |\lambda_r| < 1.} \tag{7.2}$$

Occupations and coherences can decay at different rates. Actual readings also depend on which modes the initial state excites and which components the observable can access. Internal composition can continue changing after mean energy has approached equilibrium. Under weak exchange, removing the net cycle phase on the restored family of states yields continuous relaxation equations from the slow-time limit of the same cycle, with errors controlled over finite slow times. Exact cycle factors and continuous rates thus have a common dynamical origin.

Cooling is calculated from this actual process as well. System energy changes include both thermal exchange and background restoration:

$$\boxed{E_{n+1} - E_n = Q_T(n) + Q_R(n).} \tag{7.3}$$

As the environment temperature changes, whether the system remains close to a thermal state depends on the competition between target-distribution motion and its own relaxation. For the zero-temperature exit process with fixed cycle data considered here, nonzero energy decays exponentially at late times according to the actual slow energy mode. Temperature calibrated from the full finite spectrum then satisfies $T_E \sim \Delta/(\Gamma_E t)$, where $\Delta = \log p$ is the first excitation gap. Energy-based temperature calibrates only mean energy; a full thermal state must still be assessed through the distribution and coherence.

The time direction of thermalization is linked to the fate of the outputs. Returning to fixed finite temperature and same-temperature restoration, the decrease in deviation from the thermal state gives the relaxation entropy production. Including the change in the actual pure restoration preparation yields

$$\boxed{\begin{aligned} \Sigma_n &= D_n - D_{n+1} \\ &= \Delta S_S - \beta(Q_T + Q_R) \geq 0, \\ \Sigma_n + C_R &= I_T + D(\eta'_T \| \eta_T) + I_R. \end{aligned}} \tag{7.4}$$

The first relation determines the system's approach to equilibrium. The second balances correlations, thermal-environment changes, and restoration preparation in the joint realization used here. At system equilibrium, $\Sigma_n = 0$, yet auxiliaries can still develop correlations and change their preparations. Any added label-copying or storage devices must also be included in the joint balance.

Retaining the complete final joint state and its correlations and applying the inverse joint unitaries in reverse order still recovers the initial state. Gradual loss of initial-state memory in the system is compatible with reversible joint evolution. If old auxiliaries participate again, subsequent exchange must be calculated from the actual joint state and its correlations. The complete action specified in the main text gives a return curve distinct from the fresh-auxiliary process.

Once channel weights, exchange angles, environment preparations, and total elapsed duration are independently fixed, equilibrium occupations, internal composition, coherence envelopes, and energy outputs must satisfy the same set of predictions. Changing how auxiliaries return further tests the role of environmental conditions. The endpoint, rate, and time direction of thermalization are thereby mutually constrained results of one dynamical process.

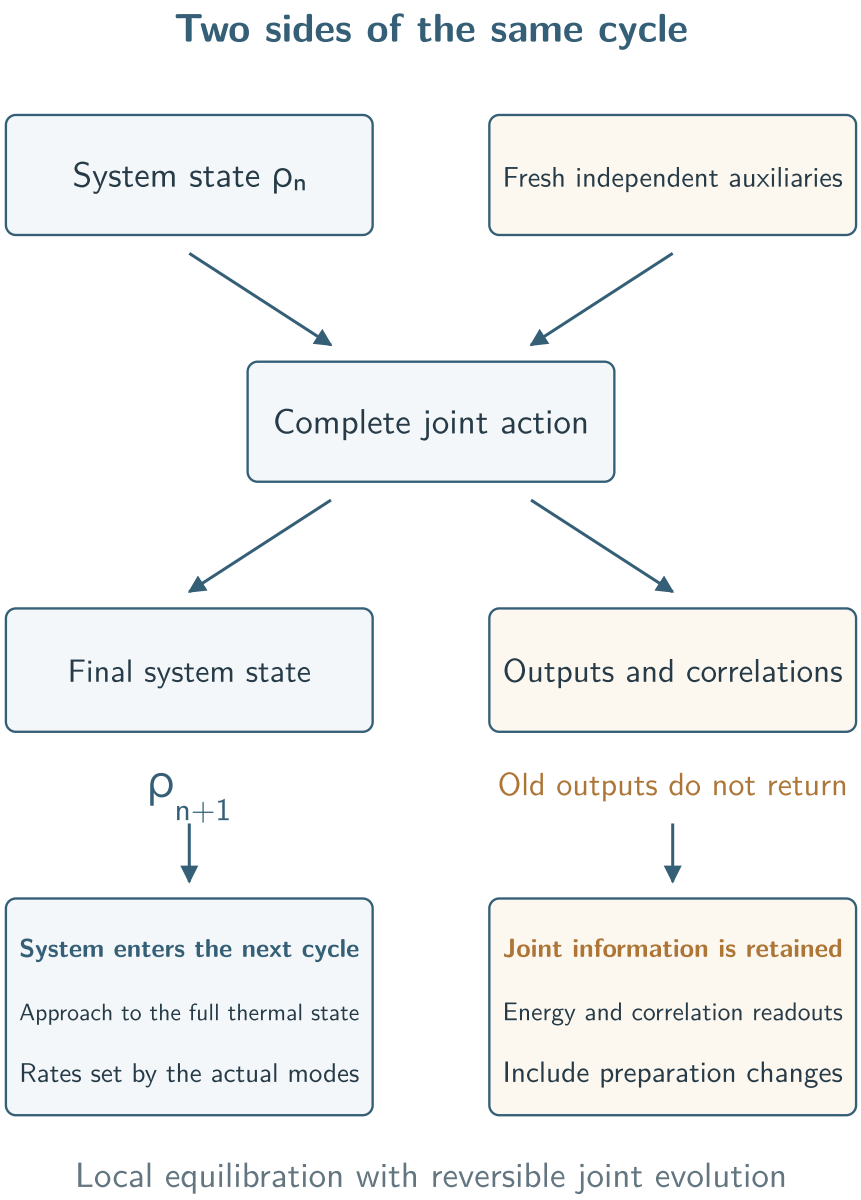


Figure 7-1. **System evolution and environment outputs in the same cycle.** The final system state enters the next cycle, while prescribed fresh auxiliaries continue to be supplied. Old outputs and correlations remain in the joint description and are treated under the no-return condition. Both marginals come from the same final joint state; showing them separately does not imply independence.

Computations and Data

The results are obtained through analytical derivations and finite-dimensional numerical checks; no experimental data are used. Curves are calculated with the preparations, channels, and initial states specified in their captions, while schematic diagrams do not represent numerical results. The accompanying programs check the complete quantum channel, relaxation spectrum, weak-exchange slow-time limit, thermal-state tracking under changing temperature, cooling corrections, energy and entropy balances, and joint evolution with fresh or returning auxiliaries. Numerical checks verify the implementation of formulas and figures. The scope of the analytical conclusions is determined by the conditions in the main text.

References

[1] S.-W. Hu, *Dynamics of the Bost–Connes State Space*, Zenodo, version 2 (2026). DOI: 10.5281/zenodo.23005935.

[2] S.-W. Hu, *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System*, Zenodo, version 7 (2026). DOI: 10.5281/zenodo.23392111.

[3] J. Rau, *Relaxation Phenomena in Spin and Harmonic Oscillator Systems*, Physical Review **129**, 1880–1888 (1963). DOI: 10.1103/PhysRev.129.1880.

[4] V. Scarani, M. Ziman, P. Štelmachovič, N. Gisin and V. Bužek, *Thermalizing Quantum Machines: Dissipation and Entanglement*, Physical Review Letters **88**, 097905 (2002). DOI: 10.1103/PhysRevLett.88.097905.

[5] L. Bruneau, A. Joye and M. Merkli, *Asymptotics of repeated interaction quantum systems*, Journal of Functional Analysis **239**, 310–344 (2006). DOI: 10.1016/j.jfa.2006.02.006.

[6] M. Raginsky, *Strictly contractive quantum channels and physically realizable quantum computers*, Physical Review A **65**, 032306 (2002). DOI: 10.1103/PhysRevA.65.032306.

[7] M. Ziman, P. Štelmachovič, V. Bužek, M. Hillery, V. Scarani and N. Gisin, *Diluting quantum information: An analysis of information transfer in system-reservoir interactions*, Physical Review A **65**, 042105 (2002). DOI: 10.1103/PhysRevA.65.042105.

[8] K. Temme, M. J. Kastoryano, M. B. Ruskai, M. M. Wolf and F. Verstraete, *The χ^2-divergence and mixing times of quantum Markov processes*, Journal of Mathematical Physics **51**, 122201 (2010). DOI: 10.1063/1.3511335.

[9] M. Lostaglio, K. Korzekwa, D. Jennings and T. Rudolph, *Quantum Coherence, Time-Translation Symmetry, and Thermodynamics*, Physical Review X **5**, 021001 (2015). DOI: 10.1103/PhysRevX.5.021001.

[10] S. Attal and Y. Pautrat, *From Repeated to Continuous Quantum Interactions*, Annales Henri Poincaré **7**, 59–104 (2006). DOI: 10.1007/s00023-005-0242-8.

[11] S. Attal and A. Joye, *Weak Coupling and Continuous Limits for Repeated Quantum Interactions*, Journal of Statistical Physics **126**, 1241–1283 (2007). DOI: 10.1007/s10955-006-9085-z.

[12] V. Gorini, A. Kossakowski and E. C. G. Sudarshan, *Completely positive dynamical semigroups of N-level systems*, Journal of Mathematical Physics **17**, 821–825 (1976). DOI: 10.1063/1.522979.

[13] G. Lindblad, *On the generators of quantum dynamical semigroups*, Communications in Mathematical Physics **48**, 119–130 (1976). DOI: 10.1007/BF01608499.

[14] K. Macieszczak, M. Guţă, I. Lesanovsky and J. P. Garrahan, *Towards a Theory of Metastability in Open Quantum Dynamics*, Physical Review Letters **116**, 240404 (2016). DOI: 10.1103/PhysRevLett.116.240404.

[15] J. E. Avron, M. Fraas, G. M. Graf and P. Grech, *Adiabatic Theorems for Generators of Contracting Evolutions*, Communications in Mathematical Physics **314**, 163–191 (2012). DOI: 10.1007/s00220-012-1504-1.

[16] L. Masanes and J. Oppenheim, *A general derivation and quantification of the third law of thermodynamics*, Nature Communications **8**, 14538 (2017). DOI: 10.1038/ncomms14538.

[17] W. H. Zurek, *Pointer basis of quantum apparatus: Into what mixture does the wave packet collapse?*, Physical Review D **24**, 1516 (1981). DOI: 10.1103/PhysRevD.24.1516.

[18] H. Umegaki, *Conditional expectation in an operator algebra, IV (Entropy and information)*, Kodai Mathematical Seminar Reports **14**, 59–85 (1962). DOI: 10.2996/kmj/1138844604.

[19] G. Lindblad, *Completely positive maps and entropy inequalities*, Communications in Mathematical Physics **40**, 147–151 (1975). DOI: 10.1007/BF01609396.

[20] M. Esposito, K. Lindenberg and C. Van den Broeck, *Entropy production as correlation between system and reservoir*, New Journal of Physics **12**, 013013 (2010). DOI: 10.1088/1367-2630/12/1/013013.

[21] D. Reeb and M. M. Wolf, *An improved Landauer principle with finite-size corrections*, New Journal of Physics **16**, 103011 (2014). DOI: 10.1088/1367-2630/16/10/103011.

[22] R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM Journal of Research and Development **5**, 183–191 (1961). DOI: 10.1147/rd.53.0183.

[23] C. Jarzynski and D. K. Wójcik, *Classical and Quantum Fluctuation Theorems for Heat Exchange*, Physical Review Letters **92**, 230602 (2004). DOI: 10.1103/PhysRevLett.92.230602.

[24] G. E. Crooks, *Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences*, Physical Review E **60**, 2721–2726 (1999). DOI: 10.1103/PhysRevE.60.2721.

[25] U. Seifert, *Entropy Production along a Stochastic Trajectory and an Integral Fluctuation Theorem*, Physical Review Letters **95**, 040602 (2005). DOI: 10.1103/PhysRevLett.95.040602.

[26] H.-P. Breuer, E.-M. Laine and J. Piilo, *Measure for the Degree of Non-Markovian Behavior of Quantum Processes in Open Systems*, Physical Review Letters **103**, 210401 (2009). DOI: 10.1103/PhysRevLett.103.210401.

[27] E.-M. Laine, J. Piilo and H.-P. Breuer, *Witness for initial system-environment correlations in open system dynamics*, EPL **92**, 60010 (2010). DOI: 10.1209/0295-5075/92/60010.

[28] N. K. Bernardes, A. R. R. Carvalho, C. H. Monken and M. F. Santos, *Environmental correlations and Markovian to non-Markovian transitions in collisional models*, Physical Review A **90**, 032111 (2014). DOI: 10.1103/PhysRevA.90.032111.

[29] N. K. Bernardes, A. Cuevas, A. Orieux, C. H. Monken, P. Mataloni, F. Sciarrino and M. F. Santos, *Experimental observation of weak non-Markovianity*, Scientific Reports **5**, 17520 (2015). DOI: 10.1038/srep17520.